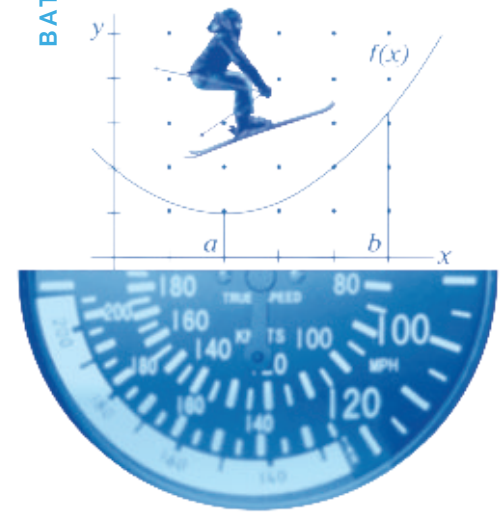




BATXILLERAT

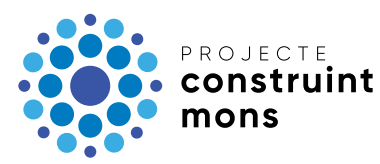
2 Matemàtiques II

SOLUCIONARI



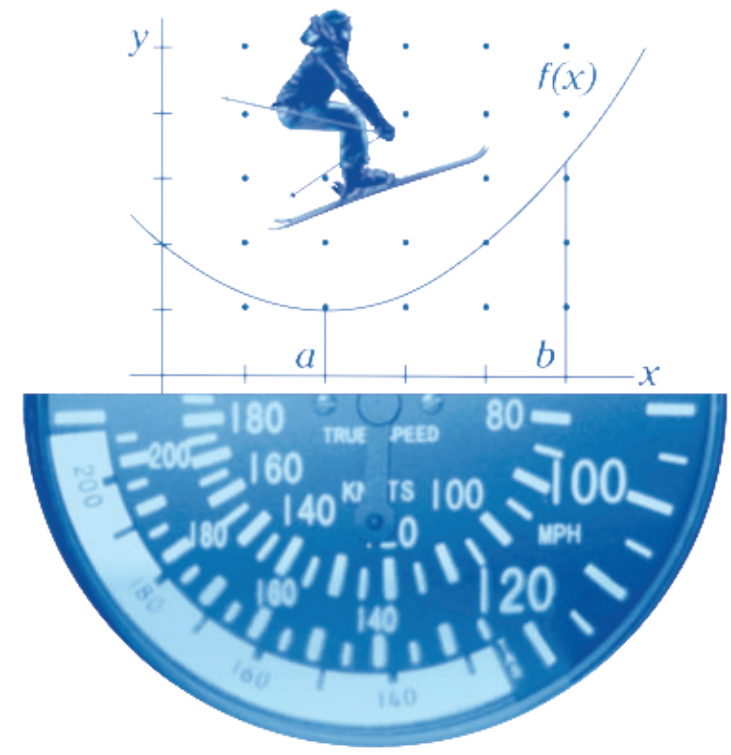
Santillana vol contribuir a la construcció d'un món més sostenible. Per això, utilitzem:

 Paper de boscos sostenibles	 Tallers amb certificació de bona gestió ambiental i energètica	 Plàstic 100 % reciclable
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Matemàtiques II

SOLUCIONARI



Matemàtiques II

SOLUCIONARI

Aquest material és una obra col·lectiva concebuda, dissenyada i creada al Departament d'Edicions de Santillana, sota la direcció de **Teresa Grence Ruiz**.

En l'elaboració ha participat:

Sonia Alejo Sánchez

María Arribas Fernández

Clara Inés Lavado Campos

Silvia Marín García

María Rosa Puebla Sánchez

Lorena Ramos San Millán

María de las Mercedes Sánchez Martín

EDICIÓ

Sonia Alejo Sánchez

José Antonio Almodóvar Herráiz

M. Àngels Andrés Casamiquela

Núria Grinyó Martorell

Clara Inés Lavado Campos

Pere Macià Arqué

Silvia Marín García

EDICIÓ EXECUTIVA

Carlos Pérez Saavedra

DIRECCIÓ DEL PROJECTE

Domingo Sánchez Figueroa



PENSA

PÀG. 267. Calcula: $\int dy$

$$\int dy = y + k$$

PÀG. 268. Existeix una funció $f(x)$ tal que

$$\int f(x) dx = 0?$$

$$\text{Si } f(x) = 0 \rightarrow \int 0 dx = k \rightarrow \text{Si } k = 0 \rightarrow$$

$$\rightarrow \int 0 dx = 0.$$

PÀG. 271. Troba $\int e^{ax} dx$ amb a nombre real.

$$\int e^{ax} dx = \frac{1}{a} \int e^{ax} \cdot a dx = \frac{1}{a} e^{ax} + k$$

PÀG. 272. Calcula: $\int \dots \int \sin x dx$
n vegades

$$\text{Si } n = 1 \rightarrow \int \sin x dx = -\cos x + k$$

$$\text{Si } n = 2 \rightarrow \int \int \sin x dx = \int -\cos x dx = -\sin x + k$$

$$\text{Si } n = 3 \rightarrow \int \int \int \sin x dx = \int -\sin x = \cos x + k$$

$$\text{Si } n = 4 \rightarrow \int \int \int \int \sin x dx = \int \cos x = \sin x + k$$

$$\text{Si } n = 4k - 3 \text{ amb } k \in \mathbb{N} \rightarrow \rightarrow \text{La integral val } -\cos x + k.$$

$$\text{Si } n = 4k - 2 \text{ amb } k \in \mathbb{N} \rightarrow \rightarrow \text{La integral val } -\sin x + k.$$

$$\text{Si } n = 4k - 1 \text{ amb } k \in \mathbb{N} \rightarrow \rightarrow \text{La integral val } \cos x + k.$$

$$\text{Si } n = 4k \text{ amb } k \in \mathbb{N} \rightarrow \rightarrow \text{La integral val } \sin x + k.$$

PÀG. 276. Calcula la integral. $\int \frac{\sqrt[4]{a+x}}{(x+a)^3} dx$

$$\begin{aligned} \int (a+x)^{\frac{1}{4}} (a+x)^{-3} dx &= \int (a+x)^{-\frac{11}{4}} dx = \\ &= \frac{(a+x)^{-\frac{7}{4}}}{-\frac{7}{4}} = -\frac{4}{7(a+x)^{\frac{7}{4}}} + k \end{aligned}$$

ACTIVITATS

- 1 Determina si la funció $F(x) = x^4$ és una funció primitiva de $f(x) = 4x^3$. Escribe dues funcions que també siguin primitives d'aquesta funció.

$$F(x) = x^4 \rightarrow F'(x) = 4x^3 \rightarrow F(x) \text{ és una funció primitiva de } f(x).$$

Resposta oberta. Per exemple:

$$F(x) = x^4 - 3$$

$$F(x) = x^4 + 1$$

- 2 Determina una primitiva de les funcions.

$$\text{a) } f(x) = 5x^4 \quad \text{b) } f(x) = 2e^x \quad \text{c) } f(x) = \frac{x^3}{2}$$

$$\text{a) } f(x) = 5x^4 \rightarrow F(x) = x^5 + k$$

$$\text{b) } f(x) = 2e^x \rightarrow F(x) = 2e^x + k$$

$$\text{c) } f(x) = \frac{x^3}{2} \rightarrow F(x) = \frac{x^4}{8} + k$$

- 3 Resol les integrals següents.

$$\text{a) } \int (5x^4 + 3x^2) dx \quad \text{c) } \int (-\cos x + 3x) dx$$

$$\text{b) } \int (4 \cos x + 2x) dx \quad \text{d) } \int (x^3 + 2) dx$$

$$\text{a) } \int (5x^4 + 3x^2) dx = x^5 + x^3 + k$$

$$\text{b) } \int (4 \cos x + 2x) dx = 4 \sin x + x^2 + k$$

$$\text{c) } \int (-\cos x + 3x) dx = -\sin x + \frac{3x^2}{2} + k$$

$$\text{d) } \int (x^3 + 2) dx = \frac{x^4}{4} + 2x + k$$

- 4 Si $\int f(x) dx = F(x) + a$ i $\int g(x) dx = G(x) + b$, troba:

$$\text{a) } \int [f(x) + g(x)] dx$$

$$\text{b) } \int [2f(x) - g(x)] dx$$

$$\text{c) } \int \left[\frac{1}{2} f(x) - 2g(x) \right] dx$$

$$\text{d) } \int [-f(x) + b \cdot g(x)] dx$$

$$a) \int [f(x) + g(x)] dx = F(x) + G(x) + k$$

$$b) \int [2f(x) - g(x)] dx = 2F(x) - G(x) + k$$

$$c) \int \left[\frac{1}{2} f(x) - 2g(x) \right] dx = \frac{1}{2} F(x) - 2G(x) + k$$

$$d) \int [-f(x) + b \cdot g(x)] dx = -F(x) + b \cdot G(x) + k$$

- 5 Troba el resultat d'aquestes integrals i escriu-les en forma de potència.

$$a) \int x^2 dx \quad b) \int \frac{2}{x^3} dx \quad c) \int 2\sqrt[3]{x^2} dx$$

$$a) \int x^2 dx = x^3 + k$$

$$b) \int \frac{2}{x^3} dx = -\frac{4}{x^2} + k$$

$$c) \int 2\sqrt[3]{x^2} dx = \frac{6\sqrt[3]{x^5}}{5} + k$$

- 6 Calcula el resultat de les integrals potencials següents.

$$a) \int -4(3 - 2x^2) dx \quad b) \int 2x\sqrt{x^2 - 1} dx$$

$$a) \int -4(3 - 2x^2) dx = -4 \int (3 - 2x^2) dx = -4 \left(3x - \frac{2x^3}{3} \right) + k$$

$$b) \int 2x\sqrt{x^2 - 1} dx = \frac{2\sqrt{(x^2 - 1)^3}}{3} + k$$

- 7 Calcula aquestes integrals de funcions potencials.

$$a) \int x(1 + 2x^2) dx \quad b) \int x^2(2 - 2x^3) dx$$

$$a) \int x(1 + 2x^2) dx = \frac{1}{4} \int 4x(1 + 2x^2) dx = \frac{1}{4} \cdot \frac{1}{2} (1 + 2x^2)^2 + k = \frac{1}{8} (1 + 2x^2)^2 + k$$

$$b) \int x^2(2 - 2x^3) dx =$$

$$= -\frac{1}{6} \int -6x^2(2 - 2x^3) dx =$$

$$= -\frac{1}{6} \cdot \frac{1}{2} (2 - 2x^3)^2 + k =$$

$$= -\frac{1}{12} (2 - 2x^3)^2 + k$$

- 8 Troba les integrals següents.

$$a) \int \sqrt{7 + 2x} dx \quad b) \int (x^2 + 1)^2 dx$$

$$a) \int \sqrt{7 + 2x} dx = \frac{1}{2} \int 2\sqrt{7 + 2x} dx = \frac{1}{2} \cdot \frac{3}{2} \sqrt{(7 + 2x)^3} + k = \frac{3}{4} \sqrt{(7 + 2x)^3} + k$$

$$b) \int (x^2 + 1)^2 dx = \int (x^4 + 2x^2 + 1) dx = \frac{x^5}{5} + \frac{2x^3}{3} + x + k$$

- 9 Calcula integrals de tipus logarítmic.

$$a) \int \frac{2x}{x^2 + 3} dx \quad b) \int \frac{2}{x - 3} dx$$

$$a) \int \frac{2x}{x^2 + 3} dx = \ln |x^2 + 3| + k$$

$$b) \int \frac{2}{x - 3} dx = 2 \ln |x - 3| + k$$

- 10 Resol les integrals següents.

$$a) \int \frac{x}{(3x^2 - 2)^3} dx \quad b) \int \frac{x}{\sqrt[3]{2 - x^2}} dx$$

$$a) \int \frac{x}{(3x^2 - 2)^3} dx = \frac{1}{6} \int \frac{6x}{(3x^2 - 2)^3} dx = \frac{1}{6} \cdot \frac{1}{-2(3x^2 - 2)^2} + k = -\frac{1}{12(3x^2 - 2)^2} + k$$

$$\begin{aligned}
 \text{b) } \int \frac{x}{\sqrt[3]{2-x^2}} dx &= \\
 &= -\frac{1}{2} \int \frac{-2x}{\sqrt[3]{2-x^2}} dx = \\
 &= \frac{1}{2} \cdot \frac{3}{2} \sqrt[3]{(2-x^2)^2} + k = \\
 &= \frac{3}{4} \sqrt[3]{(2-x^2)^2} + k
 \end{aligned}$$

11 Calcula les integrals següents.

$$\begin{array}{ll}
 \text{a) } \int 3^{\frac{x}{2}} dx & \text{c) } \int \left(\frac{1}{2}\right)^{4x} dx \\
 \text{b) } \int e^{x+1} dx & \text{d) } \int (e^{-3x} + e^{x-2}) dx
 \end{array}$$

$$\begin{aligned}
 \text{a) } \int 3^{\frac{x}{2}} dx &= 2 \int \frac{1}{2} 3^{\frac{x}{2}} dx = 2 \cdot \frac{3^{\frac{x}{2}}}{\ln 3} + k = \\
 &= \frac{2 \cdot 3^{\frac{x}{2}}}{\ln 3} + k
 \end{aligned}$$

$$\text{b) } \int e^{x+1} dx = e^{x+1} + k$$

$$\begin{aligned}
 \text{c) } \int \left(\frac{1}{2}\right)^{4x} dx &= \frac{1}{4} \int 4 \left(\frac{1}{2}\right)^{4x} dx = \\
 &= \frac{1}{4} \cdot \frac{\left(\frac{1}{2}\right)^{4x}}{\ln \frac{1}{2}} + k = -\frac{\left(\frac{1}{2}\right)^{4x}}{4 \ln 2} + k
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \int (e^{-3x} + e^{x-2}) dx &= \\
 &= \int e^{-3x} dx + \int e^{x-2} dx = \\
 &= -\frac{e^{-3x}}{3} + e^{x-2} + k
 \end{aligned}$$

12 Troba aquestes integrals de funcions exponencials.

$$\begin{array}{ll}
 \text{a) } \int 7^{x^2-1} \cdot 2x dx & \text{c) } \int \frac{3^{5x-1}}{7} dx \\
 \text{b) } \int 5e^{\frac{x}{2}+2} dx & \text{d) } \int \frac{x}{e^{x^3}} dx
 \end{array}$$

$$\text{a) } \int 7^{x^2+1} \cdot 2x dx = \frac{7^{x^2+1}}{\ln 7} + k$$

$$\text{b) } \int 5e^{\frac{x}{2}+2} dx = 10 \int \frac{1}{2} e^{\frac{x}{2}+2} dx =$$

$$= 10e^{\frac{x}{2}+2} + k$$

$$\begin{aligned}
 \text{c) } \int \frac{3^{5x-1}}{7} dx &= \frac{1}{35} \int 5 \cdot 3^{5x-1} dx = \\
 &= \frac{1}{35} \cdot \frac{3^{5x-1}}{\ln 3} + k = \frac{3^{5x-1}}{35 \ln 3} + k
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \int \frac{x}{e^{x^2}} dx &= \int x e^{-x^2} dx = \\
 &= \frac{-1}{2} \int -2x e^{-x^2} dx = \frac{-1}{2} e^{-x^2} + k = \\
 &= -\frac{1}{2e^{x^2}} + k
 \end{aligned}$$

13 Calcula aquestes integrals de funcions trigonomètriques.

$$\begin{array}{ll}
 \text{a) } \int \sin(2x) dx & \text{c) } \int \frac{\sin \frac{x}{2}}{2} dx
 \end{array}$$

$$\begin{array}{ll}
 \text{b) } \int \cos(x+1) dx & \text{d) } \int \sin(-x) dx
 \end{array}$$

$$\begin{aligned}
 \text{a) } \int \sin(2x) dx &= \frac{1}{2} \int 2 \sin(2x) dx = \\
 &= -\frac{\cos(2x)}{2} + k
 \end{aligned}$$

$$\text{b) } \int \cos(x+1) dx = \sin(x+1) + k$$

$$\begin{aligned}
 \text{c) } \int \frac{\sin \frac{x}{2}}{2} dx &= \int \frac{1}{2} \sin \frac{x}{2} dx = \\
 &= -\cos \frac{x}{2} + k
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \int \sin(-x) dx &= -\int -\sin(-x) dx = \\
 &= \cos(-x) + k
 \end{aligned}$$

14 Troba les integrals següents.

$$\text{a) } \int \frac{1}{\cos^2(x+1)} dx$$

$$\text{b) } \int -3 \sin(2x+1) dx$$

$$\text{c) } \int (x+1) \cos(x^2+2x) dx$$

$$\text{d) } \int \frac{x}{\cos^2(x^2-3)} dx$$

$$a) \int \frac{1}{\cos^2(x+1)} dx = \operatorname{tg}(x+1) + k$$

$$b) \int -3\sin(2x+1) dx = \\ = \frac{-3}{2} \int 2\sin(2x+1) dx = \\ = \frac{3}{2} \cos(2x+1) + k$$

$$c) \int (x+1)\cos(x^2+2x) dx = \\ = \frac{1}{2} \int 2(x+1) \cdot \cos(x^2+2x) dx = \\ = \frac{1}{2} \sin(x^2+2x) + k$$

$$d) \int \frac{x}{\cos^2(x^2-3)} dx = \\ = \frac{1}{2} \int \frac{2x}{\cos^2(x^2-3)} dx = \\ = \frac{1}{2} \operatorname{tg}(x^2-3) + k$$

15 Resol aquestes integrals de tipus funcions arc.

$$a) \int \frac{1}{\sqrt{1-25x^2}} dx \quad b) \int \frac{1}{1+(x-3)^2} dx$$

$$a) \int \frac{1}{\sqrt{1-25x^2}} dx = \\ = \frac{1}{5} \int \frac{5}{\sqrt{1-25x^2}} dx = \arcsin(5x) + k$$

$$b) \int \frac{1}{1+(x-3)^2} dx = \operatorname{arctg}(x-3) + k$$

16 Troba la solució de les integrals següents.

$$a) \int \frac{1}{\sqrt{1-(2x-3)^2}} dx \quad b) \int \frac{x}{1+9x^4} dx$$

$$a) \int \frac{1}{\sqrt{1-(2x-3)^2}} dx = \\ = \frac{1}{2} \int \frac{2}{\sqrt{1-(2x-3)^2}} dx = \\ = \frac{1}{2} \arcsin(2x-3) + k$$

$$b) \int \frac{x}{1+9x^4} dx = \frac{1}{6} \int \frac{6x}{1+(3x^2)^2} dx =$$

$$= \frac{1}{6} \operatorname{arctg}(3x^2) + k$$

17 Troba les integrals integrant per parts.

$$a) \int x^2 \ln x dx$$

$$b) \int x^2 \sin x dx$$

$$u = \ln x \rightarrow du = \frac{1}{x} dx \\ dv = x^2 dx \rightarrow v = \frac{x^3}{3}$$

$$a) \int x^2 \ln x dx = \frac{x^3}{3} \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} dx = \\ = \frac{x^3}{3} \ln x - \int \frac{x^2}{3} dx = \\ = \frac{x^3}{3} \ln x - \frac{x^3}{9} + k$$

$$u = x^2 \rightarrow du = 2x dx \\ dv = \sin x dx \rightarrow v = -\cos x$$

$$b) \int x^2 \sin x dx = -x^2 \cos x +$$

$$u = 2x \rightarrow du = 2 dx \\ dv = \cos x dx \rightarrow v = \sin x$$

$$+ \int 2x \cos x dx = -x^2 \cos x + \\ + 2x \sin x - \int 2 \sin x dx = \\ = -x^2 \cos x + 2x \sin x + 2 \cos x + k$$

18 Utilitza la integració per parts per calcular:

$$a) \int e^x (7+2x) dx$$

$$b) \int (4x^2 - 3x + 1) \cos x dx$$

$$u = 7+2x \rightarrow du = 2 dx \\ dv = e^x dx \rightarrow v = e^x$$

$$a) \int e^x (7+2x) dx = (7+2x)e^x - \\ - \int 2e^x dx = (7+2x)e^x - 2e^x + k = \\ = e^x (5+2x) + k$$

$$u = 4x^2 - 3x + 1 \rightarrow du = (8x - 3) dx$$

$$dv = \sin x dx \rightarrow v = -\cos x$$

$$b) \int (4x^2 - 3x + 1) \sin x dx =$$

$$= -(4x^2 - 3x + 1) \cos x +$$

$$+ \int (8x - 3) \cos x dx =$$

$$u = 8x - 3 \rightarrow du = 8 dx$$

$$dv = \cos x dx \rightarrow v = \sin x$$

$$= -(4x^2 - 3x + 1) \cos x +$$

$$+ (8x - 3) \sin x - \int 8 \sin x dx =$$

$$= -(4x^2 - 3x + 1) \cos x +$$

$$+ (8x - 3) \sin x + 8 \cos x + k$$

19 Resol aquestes integrals racionals.

$$a) \int \frac{2}{x^2 - 1} dx$$

$$b) \int -\frac{3}{x^2 + x - 2} dx$$

$$a) \int \frac{2}{x^2 - 1} dx =$$

$$\int \frac{-1}{x + 1} dx + \int \frac{1}{x - 1} dx =$$

$$= -\ln|x + 1| + \ln|x - 1| + k$$

$$b) \int -\frac{3}{x^2 + x - 2} dx =$$

$$= \int \frac{1}{x + 2} dx + \int \frac{-1}{x - 1} dx =$$

$$= \ln|x + 2| - \ln|x - 1| + k$$

20 Calcula les integrals racionals següents.

$$a) \int \frac{2x + 1}{x^4 - 5x^2 + 4} dx$$

$$b) \int \frac{7x - 2}{x^3 - 2x^2 - x + 2} dx$$

$$a) \int \frac{2x + 1}{x^4 - 5x^2 + 4} dx =$$

$$= \frac{5}{12} \int \frac{1}{x - 2} dx - \frac{1}{2} \int \frac{1}{x - 1} dx -$$

$$- \frac{1}{6} \int \frac{1}{x + 1} dx + \frac{1}{4} \int \frac{1}{x + 2} dx =$$

$$= \frac{5}{12} \ln|x - 2| - \frac{1}{2} \ln|x - 1| -$$

$$- \frac{1}{6} \ln|x + 1| + \frac{1}{4} \ln|x + 2| + k$$

$$b) \int \frac{7x - 2}{x^3 - 2x^2 - x + 2} dx =$$

$$= 4 \int \frac{1}{x - 2} dx - \frac{5}{2} \int \frac{1}{x - 1} dx -$$

$$- \frac{3}{2} \int \frac{1}{x + 1} dx =$$

$$= 4 \ln|x - 2| - \frac{5}{2} \ln|x - 1| -$$

$$- \frac{3}{2} \ln|x + 1| + k$$

21 Calcula aquestes integrals racionals.

$$a) \int \frac{x^2}{(x - 1)^3} dx$$

$$b) \int -\frac{3x - 2}{(2 - x)^2} dx$$

$$a) \int \frac{x^2}{(x - 1)^3} dx =$$

$$= \int \frac{1}{(x - 1)^3} dx + 2 \int \frac{1}{(x - 1)^2} dx +$$

$$+ \int \frac{1}{x - 1} dx =$$

$$= \frac{-1}{2(x - 1)^2} - \frac{2}{x - 1} + \ln|x - 1| + k$$

$$b) \int -\frac{3x - 2}{(2 - x)^2} dx =$$

$$= 4 \int \frac{-1}{(2 - x)^2} dx - 3 \int \frac{-1}{2 - x} dx =$$

$$= \frac{-4}{2 - x} - 3 \ln|2 - x| + k$$

22 Resol les integrals racionals següents.

$$a) \int \frac{-2x^2 + 1}{x^3 + 6x^2 + 12x + 8} dx$$

$$b) \int \frac{x - 2}{x^4} dx$$

$$\begin{aligned} \text{a) } \int \frac{-2x^2 + 1}{x^3 + 6x^2 + 12x + 8} dx &= \\ &= -7 \int \frac{1}{(x+2)^3} dx + 8 \int \frac{1}{(x+2)^2} dx - \\ &\quad - 2 \int \frac{1}{x+2} dx = \\ &= \frac{7}{2(x+2)^2} - \frac{8}{x+2} - 2 \ln|x+2| + k \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{x-2}{x^4} dx &= \\ &= \int \frac{1}{x^3} dx - 2 \int \frac{1}{x^4} dx = \\ &= -\frac{1}{2x^2} + \frac{2}{3x^3} + k \end{aligned}$$

23 Resol aquesta integral racional.

$$\begin{aligned} &\int \frac{4x^2 - 2x}{(x+2)(x-3)^2} dx \\ &\int \frac{4x^2 - 2x}{(x+2)(x-3)^2} dx = \\ &= 6 \int \frac{1}{(x-3)^2} dx + \frac{16}{5} \int \frac{1}{x-3} dx + \\ &\quad + \frac{4}{5} \int \frac{1}{x+2} dx = -\frac{6}{x-3} + \\ &\quad + \frac{16}{5} \ln|x-3| + \frac{4}{5} \ln|x+2| + k \end{aligned}$$

24 Calcula la integral racional següent.

$$\begin{aligned} &\int \frac{-x^2 + 7x}{x^3 - x^2 - x + 1} dx \\ &\int \frac{-x^2 + 7x}{x^3 - x^2 - x + 1} dx = \\ &= 3 \int \frac{1}{(x-1)^2} dx + \int \frac{1}{x-1} dx - \\ &\quad - 2 \int \frac{1}{x+1} dx = \\ &= -\frac{3}{x-1} + \ln|x-1| - 2 \ln|x+1| + k \end{aligned}$$

25 Resol aquestes integrals racionals.

$$\text{a) } \int \frac{2}{x^2 + 1} dx \quad \text{b) } \int -\frac{3x-2}{2+x^2} dx$$

$$\text{a) } \int \frac{2}{x^2 + 1} dx = 2 \operatorname{arctg} x + k$$

$$\begin{aligned} \text{b) } \int -\frac{3x-2}{2+x^2} dx &= -\frac{3}{2} \ln|2+x^2| + \\ &\quad + \frac{2}{\sqrt{2}} \operatorname{arctg}\left(\frac{x}{\sqrt{2}}\right) + k \end{aligned}$$

26 Calcula les integrals racionals següents.

$$\text{a) } \int \frac{-2x^2 + 1}{x^3 - x^2 + 3x - 3} dx$$

$$\text{b) } \int \frac{x-2}{x^2(x^2+1)} dx$$

$$\begin{aligned} \text{a) } \int \frac{-2x^2 + 1}{x^3 - x^2 + 3x - 3} dx &= \\ &= \int \frac{-1}{x-1} dx + \int \frac{-\frac{7}{4}x - \frac{7}{4}}{x^2 + 3} dx = \\ &= \int \frac{-1}{x-1} dx + \int \frac{-\frac{7}{4}x}{x^2 + 3} dx + \\ &\quad + \int \frac{-\frac{7}{4}}{x^2 + 3} dx = \\ &= -\frac{1}{4} \ln|x-1| - \frac{7}{8} \ln|3+x^2| - \\ &\quad - \frac{7}{4\sqrt{3}} \operatorname{arctg}\left(\frac{x}{\sqrt{3}}\right) + k \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{x-2}{x^2(x^2+1)} dx &= \\ &= \int \frac{-2}{x^2} dx + \int \frac{1}{x} dx + \\ &\quad + \int \frac{-x+2}{1+x^2} dx = \\ &= \int \frac{-2}{x^2} dx + \int \frac{1}{x} dx + \\ &\quad + \int \frac{-x}{1+x^2} dx + \int \frac{2}{1+x^2} dx = \\ &= \frac{2}{x} + \ln|x| - \frac{1}{2} \ln|1+x^2| + \\ &\quad + 2 \operatorname{arctg} x + k \end{aligned}$$

27 Resol aquestes integrals racionals.

$$\text{a) } \int \frac{2x^4}{(x-1)^3} dx$$

$$b) \int -\frac{3x^3 - 2}{(2 - x)^2} dx$$

$$a) \int \frac{2x^4}{(x - 1)^3} dx = \\ = x^2 + 6x + 12 \ln |x - 1| - \\ - \frac{8}{x - 1} - \frac{1}{(x - 1)^2} + k$$

$$b) \int -\frac{3x^3 - 2}{(2 - x)^2} dx = \\ = \frac{-3x^2}{2} - 12x + 22 \ln |2 - x| - \\ - \frac{36}{2 - x} + k$$

28 Calcula les integrals racionals següents.

$$a) \int \frac{-2x^5 + 1}{x^4 - 2x^2 + 1} dx$$

$$b) \int \frac{x^6 - 1}{x^2(x^2 + 1)(x - 1)} dx$$

$$a) \int \frac{-2x^5 + 1}{x^4 - 2x^2 + 1} dx = \\ = \int -2x dx + \int \frac{-1}{(x - 1)^2} dx + \\ + \int \frac{-9}{4(x - 1)} dx + \int \frac{3}{4(x + 1)^2} dx + \\ + \int \frac{-7}{4(x + 1)} dx = \\ = -x^2 + \frac{1}{4(x - 1)} - \frac{9}{4} \ln |x - 1| - \\ - \frac{3}{4(x + 1)} - \frac{7}{4} \ln |x + 1| + k$$

$$b) \int \frac{x^6 - 1}{x^2(x^2 + 1)(x - 1)} dx = \\ = \int \left(x + 1 + \frac{1}{x^2} + \frac{1}{x} + \frac{-x}{x^2 + 1} + \right. \\ \left. + \frac{-1}{x^2 + 1} \right) dx = \frac{x^2}{2} + x - \frac{1}{x} + \ln |x| - \\ - \frac{1}{2} \ln |x^2 + 1| - \arctg x + k$$

29 Calcula aquestes integrals mitjançant un canvi de variable.

$$a) \int x^2(7 + 2x^3) dx$$

$$b) \int \frac{\log^5 x}{x} dx$$

$$\begin{array}{l} t = 7 + 2x^3 \\ dt = 6x^2 dx \end{array}$$

$$a) \int x^2(7 + 2x^3) dx = \frac{1}{6} \int t dt = \frac{t^2}{12} + k = \\ = \frac{(7 + 2x^3)^2}{12} + k$$

$$\begin{array}{l} t = \log x \\ dt = \frac{1}{\ln 10 \cdot x} dx \end{array}$$

$$b) \int \frac{\log^5 x}{x} dx = \ln 10 \int t^5 dt = \\ = \ln 10 \cdot \frac{t^6}{6} + k = \ln 10 \cdot \frac{\log^6 x}{6} + k$$

30 Troba les integrals amb un canvi de variable.

$$a) \int e^{x^2}(7 + 2x) dx$$

$$b) \int \frac{4x}{\sqrt{2 - 3x^2}} dx$$

$$\begin{array}{l} t = x^2 \\ dt = 2x dx \end{array}$$

$$a) \int e^{x^2}(2x^3 + 2x) dx = \int (e^t t + e^t) dt = \\ = \int e^t t dt + \int e^t dt = \int e^t t dt + e^t =$$

$$\begin{array}{l} u = t \rightarrow du = dt \\ dv = e^t dt \rightarrow v = e^t \end{array}$$

$$= te^t - \int e^t dt + e^t = te^t + k = x^2 e^{x^2} + k$$

$$\begin{array}{l} t = 2 - 3x^2 \\ dt = -6x dx \end{array}$$

$$b) \int \frac{4x}{\sqrt{2 - 3x^2}} dx = \frac{-2}{3} \int \frac{1}{\sqrt{t}} dt = \\ = \frac{-4\sqrt{t}}{3} + k = \frac{-4\sqrt{2 - 3x^2}}{3} + k$$

31 Calcula aquestes integrals.

a) $\int \sin^5 x \cos^2 x \, dx$

b) $\int \sqrt{4-x^2} \, dx$

$$\begin{array}{l} t = \cos x \\ dt = -\sin x \, dx \end{array}$$

a) $\int \sin^5 x \cos^2 x \, dx = -\int (1-t^2)^2 t^2 \, dt =$
 $= -\int (t^6 - 2t^4 + t^2) \, dt =$
 $= -\frac{t^7}{7} + \frac{2t^5}{5} - \frac{t^3}{3} + k =$
 $= -\frac{\cos^7 x}{7} + \frac{2\cos^5 x}{5} - \frac{\cos^3 x}{3} + k$

$$\begin{array}{l} \frac{x}{2} = \sin t \\ dx = 2 \cos t \, dt \end{array}$$

b) $\int \sqrt{4-x^2} \, dx = 2 \int \sqrt{1-\left(\frac{x}{2}\right)^2} \, dx =$
 $= 2 \int \sqrt{1-\sin^2 t} \cdot 2 \cos t \, dt =$
 $= 4 \int \cos^2 t \, dt =$
 $= 4 \int \frac{1+\cos(2t)}{2} \, dt =$
 $= 2t + 2\sin(2t) + k =$
 $= 2\arcsin \frac{x}{2} + 2\sin\left(2\arcsin \frac{x}{2}\right) + k =$
 $= 2\arcsin \frac{x}{2} + 2 \cdot \frac{x}{2} \sqrt{1-\left(\frac{x}{2}\right)^2} + k =$
 $= 2\arcsin \frac{x}{2} + \frac{x}{2} \sqrt{4-x^2} + k$

32 Troba la solució de les integrals.

a) $\int \sin^2 x \, dx$

b) $\int \frac{\sqrt{2-x^2}}{4} \, dx$

a) $\int \sin^2 x \, dx = \int \frac{1-\cos(2x)}{2} \, dx =$
 $= \frac{x}{2} + \frac{\sin(2x)}{4} + k$

b) $\int \frac{\sqrt{2-x^2}}{4} \, dx = \frac{\frac{x}{\sqrt{2}} = \sin t}{dx = \sqrt{2} \cos t \, dt}$
 $= \frac{\sqrt{2}}{4} \int \sqrt{1-\left(\frac{x}{\sqrt{2}}\right)^2} \, dx =$
 $= \frac{\sqrt{2}}{4} \int \sqrt{1-\sin^2 t} \sqrt{2} \cos t \, dt =$
 $= \frac{1}{2} \int \cos^2 t \, dt = \frac{1}{2} \int \frac{1+\cos(2t)}{2} \, dt =$
 $= \frac{1}{4} \left(t + \frac{\sin(2t)}{2} \right) + k =$
 $= \frac{1}{4} \arcsin \frac{x}{\sqrt{2}} + \frac{x}{8} \sqrt{2-x^2} + k$

PRACTICA

33 Calcula la funció $f(x)$ la gràfica de la qual passa pel punt $(0, 1)$ (és a dir, $f(0) = 1$) i que té com a derivada la funció $f'(x) = \frac{2x}{x^2+1}$.

$$f(x) = \int \frac{2x}{x^2+1} \, dx = \ln|x^2+1| + k$$

$$f(0) = 1 \rightarrow \ln|1| + k = 1 \rightarrow k = 1 \rightarrow$$

$$\rightarrow f(x) = \ln|x^2+1| + 1$$

34 Resol la integral $\int \frac{dx}{\sqrt{4-(2x+1)^2}}$.

$$\int \frac{dx}{\sqrt{4-(2x+1)^2}} =$$

$$= \int \frac{\frac{1}{2}}{\frac{1}{2}\sqrt{4-(2x+1)^2}} \, dx =$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{1-\left(\frac{2x+1}{2}\right)^2}} \, dx =$$

$$= \frac{1}{2} \arcsin\left(\frac{2x+1}{2}\right) + k$$

35 Troba una primitiva de la integral

$$\int \frac{x^2}{\sqrt{1-x^3}} \, dx \text{ que s'anul·li en } x = 0.$$

$$F(x) = \int \frac{x^2}{\sqrt{1-x^3}} \, dx = -\frac{2}{3} \sqrt{1-x^3} + k$$

$$F(0) = 0 \rightarrow -\frac{2}{3} + k = 0 \rightarrow k = \frac{2}{3} \rightarrow$$

$$\rightarrow F(x) = -\frac{2}{3}\sqrt{1} + \frac{2}{3}$$

- 36 Utilitza el canvi de variable $t = e^{2x}$ per resoldre la integral següent: $\int e^{2x+e^{2x}} dx$.

$$t = e^{2x} \rightarrow dt = 2e^{2x} dx \rightarrow dx = \frac{dt}{2t}$$

$$\begin{aligned} \int e^{2x+e^{2x}} dx &= \int t \cdot e^t \cdot \frac{1}{2t} dt = \frac{1}{2} \int e^t dt = \\ &= \frac{1}{2} e^t + k = \frac{1}{2} e^{e^{2x}} + k \end{aligned}$$

- 37 Calcula $\int \frac{\sqrt{25-x^2}}{5} dx$.

$$x = 5 \sin t \rightarrow dx = 5 \cos t dt$$

$$\begin{aligned} \int \frac{\sqrt{25-x^2}}{5} dx &= \int \frac{\sqrt{25-25\sin^2 t}}{5} \cdot 5 \cos t dt = \\ &= \int \frac{\sqrt{25(1-\sin^2 t)}}{5} \cdot 5 \cos t dt = \\ &= 5 \int \sqrt{1-\sin^2 t} \cos t dt = \\ &= 5 \int \sqrt{\cos^2 t} \cos t dt = 5 \int \cos^2 t dt = \\ &= 5 \int \frac{1+\cos(2t)}{2} dt = \\ &= \frac{5}{2} \int dt + 5 \int \frac{\cos(2t)}{2} dt = \\ &= \frac{5}{2} t + \frac{5}{4} \sin(2t) + k = \end{aligned}$$

$$\begin{aligned} x = 5 \sin t \rightarrow t &= \arcsin\left(\frac{x}{5}\right) \\ \sin(2t) &= 2 \sin t \cos t = 2 \left(\frac{x}{5}\right) \sqrt{1-\left(\frac{x}{5}\right)^2} = \\ &= \frac{2x}{5} \sqrt{\frac{25-x^2}{25}} = \frac{2x}{25} \sqrt{25-x^2} \end{aligned}$$

$$\begin{aligned} &= \frac{5}{2} \arcsin \frac{x}{5} + \frac{5}{4} \cdot \frac{2x}{25} \sqrt{25-x^2} + k = \\ &= \frac{5}{2} \arcsin \frac{x}{5} + \frac{x}{10} \sqrt{25-x^2} + k \end{aligned}$$

- 38 Troba la integral $\int e^x \sin x dx$.

$$\begin{aligned} u &= e^x \rightarrow du = e^x dx \\ dv &= \sin x dx \rightarrow v = -\cos x \end{aligned}$$

$$\begin{aligned} I &= \int e^x \sin x dx = \\ &= -e^x \cos x + \int \cos x e^x dx = \\ &= -e^x \cos x + e^x \sin x - \int e^x \sin x dx = \\ &= -e^x \cos x + e^x \sin x - I \end{aligned}$$

Aïllem I :

$$I = -e^x \cos x + e^x \sin x - I \rightarrow$$

$$\rightarrow I = \frac{e^x(\sin x - \cos x)}{2}$$

Llavors:

$$\int e^x \sin x dx = \frac{e^x \sin x - \cos x}{2} + k$$

- 39 Resol per parts la integral següent.

$$\int 3 \ln(2x-1) dx$$

$$\begin{aligned} u &= \ln(2x-1) \rightarrow du = \frac{2}{2x-1} dx \\ dv &= 3dx \rightarrow v = 3x \end{aligned}$$

$$\begin{aligned} \int 3 \ln(2x-1) dx &= \\ &= 3x \ln(2x-1) - \int \frac{6x}{2x-1} dx = \\ &= 3x \ln(2x-1) - \int \left(3 + \frac{3}{2x-1}\right) dx = \\ &= 3x \ln(2x-1) - 3x - \frac{3}{2} \ln|2x-1| + k \end{aligned}$$

- 40 Calcula $\int (x^2+1)e^{x-2} dx$.

$$\begin{aligned} u &= x^2+1 \rightarrow du = 2x dx \\ dv &= e^{x-2} dx \rightarrow v = e^{x-2} \end{aligned}$$

$$\int (x^2+1) \cdot e^{x-2} dx =$$

$$\begin{aligned} u &= 2x \rightarrow du = 2dx \\ dv &= e^{x-2} dx \rightarrow v = e^{x-2} \end{aligned}$$

$$= (x^2+1) \cdot e^{x-2} - \int 2xe^{x-2} dx =$$

$$\begin{aligned}
 &= (x^2 + 1) \cdot e^{x-2} - 2xe^{x-2} + \int 2e^{x-2} dx = \\
 &= (x^2 + 1) \cdot e^{x-2} - 2xe^{x-2} + 2e^{x-2} + k = \\
 &= e^{x-2}(x^2 - 2x + 3) + k
 \end{aligned}$$

- 41 Utilitza el canvi de variable $t^6 = 1 + x$ per resoldre la integral

$$\int \frac{\sqrt{x+1} + 2}{(x+1)^{2/3} - \sqrt{x+1}} dx.$$

$$t^6 = 1 + x \rightarrow 6t^5 dt = dx \rightarrow t = (1+x)^{1/6}$$

$$\begin{aligned}
 &\int \frac{\sqrt{x+1} + 2}{(x+1)^{2/3} - \sqrt{x+1}} dx = \\
 &= \int \frac{\sqrt{t^6} + 2}{(t^6)^{2/3} - \sqrt{t^6}} \cdot 6t^5 dt = \\
 &= 6 \int \frac{t^3 + 2}{t^4 - t^3} \cdot t^5 dt = \\
 &= 6 \int \frac{t^5 + 2t^2}{t - 1} dt = \\
 &= 6 \int \left(t^4 + t^3 + t^2 + 3t + 3 + \frac{3}{t-1} \right) dt = \\
 &= \frac{6}{5} t^5 + \frac{3}{2} t^4 + 2t^3 + 9t^2 + 18t + \\
 &\quad + 18 \ln|t-1| + k = \frac{6}{5} (1+x)^{5/6} + \\
 &\quad + \frac{3}{2} (1+x)^{2/3} + 2(1+x)^{1/2} + 9(1+x)^{1/3} + \\
 &\quad + 18(1+x)^{1/6} + 18 \ln|(1+x)^{1/6} - 1| + k
 \end{aligned}$$

- 42 Fes un canvi de variable i resol

la integral: $\int \frac{1+x}{1+\sqrt{x}} dx$

$$t = \sqrt{x} \rightarrow dt = \frac{1}{2\sqrt{x}} dx \rightarrow dx = 2t dt$$

$$\begin{aligned}
 &\int \frac{1+x}{1+\sqrt{x}} dx = \\
 &= \int \frac{1+t^2}{1+t} \cdot 2t dt = 2 \int \frac{t^3+t}{1+t} \cdot dt = \\
 &= 2 \int \left(t^2 - t + 2 - \frac{2}{t+1} \right) dt = \\
 &= \frac{2}{3} t^3 - t^2 + 4t - 4 \ln|t+1| + k = \\
 &= \frac{2}{3} \sqrt{x}^3 - x + 4\sqrt{x} - 4 \ln|\sqrt{x} + 1| + k
 \end{aligned}$$

ACTIVITATS FINALS

1. Calcula la primitiva d'una funció elemental

Funció primitiva d'una funció



ACTIVITATS FLAIX

- 43 Comprova, en cadascun dels casos, que la funció $F(x)$ és una primitiva de la funció $f(x)$.

a) $F(x) = 3x^5 - 2x + 1$

$f(x) = 15x^4 - 2$

b) $F(x) = (x^5 - 2)^3$

$f(x) = 15x^4(x^5 - 2)^2$

c) $F(x) = \cos^2 x - 1492$

$f(x) = -\sin 2x$

a) $F'(x) = 15x^4 - 2 \rightarrow$

$\rightarrow F(x)$ és primitiva de $f(x)$.

b) $F'(x) = 15x^4(x^5 - 2)^2 \rightarrow$

$\rightarrow F(x)$ és primitiva de $f(x)$.

c) $F'(x) = 2 \sin x \cos x = -\sin(2x) \rightarrow$

$\rightarrow F(x)$ és primitiva de $f(x)$.

- 44 Troba mentalment una primitiva de cada una de les funcions polinòmiques.

a) $f(x) = 1001$

e) $f(x) = x^2$

b) $f(x) = 2x$

f) $f(x) = 4x^3$

c) $f(x) = x$

g) $f(x) = (n+1)x^n$

d) $f(x) = 3x^2$

h) $f(x) = x^n$

a) $F(x) = 1001x + k \rightarrow F'(x) = 1001$

b) $F(x) = x^2 + k \rightarrow F'(x) = 2x$

c) $F(x) = \frac{x^2}{2} + k \rightarrow F'(x) = x$

d) $F(x) = x^3 + k \rightarrow F'(x) = 3x^2$

e) $F(x) = \frac{x^3}{3} + k \rightarrow F'(x) = x^2$

f) $F(x) = x^4 + k \rightarrow F'(x) = 4x^3$

g) $F(x) = x^{n+1} + k \rightarrow F'(x) = (n+1)x^n$

h) $F(x) = \frac{x^{n+1}}{n+1} + k \rightarrow F'(x) = x^n$

- 45 Comprova, en cadascun dels casos, que la funció $F(x)$ és una primitiva de la funció $f(x)$.

a) $F(x) = \frac{x-1}{x^2} - 3$ $f(x) = \frac{-x+2}{x^3}$
 b) $F(x) = \ln \sqrt{x^2+2x}$ $f(x) = \frac{x+1}{x^2+2x}$
 c) $F(x) = 5e^{-x^2} + 11$ $f(x) = -10x e^{-x^2}$
 d) $F(x) = \arctg \sqrt{x}$ $f(x) = \frac{1}{2\sqrt{x}(1+x)}$

a) $F(x) = \frac{x-1}{x^2} - 3 \rightarrow$
 $\rightarrow F'(x) = \frac{-x+2}{x^3} \rightarrow$
 $\rightarrow F(x)$ és primitiva de $f(x)$.
 b) $F(x) = \ln \sqrt{x^2+2x} \rightarrow$
 $\rightarrow F'(x) = \frac{x+1}{x^2+2x} \rightarrow$
 $\rightarrow F(x)$ és primitiva de $f(x)$.
 c) $F(x) = 5e^{-x^2} + 11 \rightarrow$
 $\rightarrow F'(x) = -10xe^{-x^2} \rightarrow$
 $\rightarrow F(x)$ és primitiva de $f(x)$.
 d) $F(x) = \arctg \sqrt{x} \rightarrow$
 $\rightarrow F'(x) = \frac{1}{2\sqrt{x}(1+x)} \rightarrow$
 $\rightarrow F(x)$ és primitiva de $f(x)$.

- 46 Calcula una funció primitiva $F(x)$ de cada funció que compleix la condició que s'indica.

a) $f(x) = 3x^2$ $F(0) = 1$
 b) $f(x) = \frac{4}{5x}$ $F(1) = 4$
 c) $f(x) = \sin(3x)$ $F(\pi) = -\frac{1}{3}$
 d) $f(x) = e^{2x}$ $F(0) = \frac{2}{3}$
 a) $f(x) = 3x^2 \rightarrow F(x) = x^3 + k$
 $F(0) = 1 \rightarrow k = 1 \rightarrow F(x) = x^3 + 1$
 b) $f(x) = \frac{4}{5x} \rightarrow F(x) = \frac{4}{5} \ln|x| + k$
 $F(1) = 4 \rightarrow k = 4 \rightarrow F(x) = \frac{4}{5} \ln|x| + 4$
 c) $f(x) = \sin(3x) \rightarrow F(x) = -\frac{1}{3} \cos(3x) + k$

$F(\pi) = -\frac{1}{3} \rightarrow k = -\frac{2}{3} \rightarrow$
 $\rightarrow F(x) = -\frac{1}{3} \cos(3x) - \frac{2}{3}$

d) $f(x) = e^{2x} \rightarrow F(x) = \frac{e^{2x}}{2} + k$
 $F(0) = \frac{2}{3} \rightarrow k = \frac{1}{6} \rightarrow F(x) = \frac{e^{2x}}{2} + \frac{1}{6}$

- 47 **INVENTA.** ESCRIU tres funcions $f(x)$ la primitiva de les quals compleixi:

a) $F(0) = 1$ c) $F(2) = 3$
 b) $F(1) = 2$ d) $F(3) = 4$

Resposta oberta. Per exemple:

a) $f(x) = 3x^2 \rightarrow F(x) = x^3 + k \xrightarrow{k=1} F(0) = 1$
 $f(x) = 2 \rightarrow F(x) = 2x + k \xrightarrow{k=1} F(0) = 1$
 $f(x) = 4x^3 \rightarrow F(x) = x^4 + k \xrightarrow{k=1} F(0) = 1$
 b) $f(x) = 3x^2 \rightarrow F(x) = x^3 + k \xrightarrow{k=1} F(1) = 2$
 $f(x) = 2 \rightarrow F(x) = 2x + k \xrightarrow{k=0} F(1) = 2$
 $f(x) = 4x^3 \rightarrow F(x) = x^4 + k \xrightarrow{k=1} F(1) = 2$
 c) $f(x) = 3x^2 \rightarrow$
 $\rightarrow F(x) = x^3 + k \xrightarrow{k=-5} F(2) = 3$
 $f(x) = 2 \rightarrow F(x) = 2x + k \xrightarrow{k=-1} F(2) = 3$
 $f(x) = 4x^3 \rightarrow$
 $\rightarrow F(x) = x^4 + k \xrightarrow{k=-13} F(2) = 3$
 d) $f(x) = 3x^2 \rightarrow$
 $\rightarrow F(x) = x^3 + k \xrightarrow{k=-23} F(3) = 4$
 $f(x) = 2 \rightarrow F(x) = 2x + k \xrightarrow{k=-2} F(3) = 4$
 $f(x) = 4x^3 \rightarrow$
 $\rightarrow F(x) = x^4 + k \xrightarrow{k=-77} F(3) = 4$

- 48 Troba una primitiva de cada funció i comprova-ho.

a) $f(x) = \frac{1}{2\sqrt{x}}$ d) $f(x) = \frac{1}{\sqrt{x-4}}$
 b) $f(x) = \frac{1}{\sqrt{x}}$ e) $f(x) = \frac{1}{\sqrt{3x+10}}$
 c) $f(x) = \frac{1}{\sqrt{x+1}}$ f) $f(x) = \frac{1}{\sqrt{1-x^2}}$
 a) $F(x) = \sqrt{x} + k \rightarrow F'(x) = \frac{1}{2\sqrt{x}}$
 b) $F(x) = 2\sqrt{x} + k \rightarrow F'(x) = \frac{1}{\sqrt{x}}$

$$c) F(x) = 2\sqrt{x+1} + k \rightarrow F'(x) = \frac{1}{\sqrt{x+1}}$$

$$d) F(x) = 2\sqrt{x-4} + k \rightarrow F'(x) = \frac{1}{\sqrt{x-4}}$$

$$e) F(x) = \frac{2}{3}\sqrt{3x+10} + k \rightarrow$$

$$\rightarrow F'(x) = \frac{1}{\sqrt{3x+10}}$$

$$f) F(x) = \arcsin x + k \rightarrow F'(x) = \frac{1}{\sqrt{1-x^2}}$$

- 49** Troba mentalment una primitiva de cadascuna de les funcions següents i comprova la teva resposta.

$$a) f(x) = \frac{19}{x}$$

$$d) f(x) = e^{5x+55}$$

$$b) f(x) = e^{2x}$$

$$e) f(x) = \frac{4}{2x+3}$$

$$c) f(x) = \frac{1}{19x}$$

$$f) f(x) = 2^x$$

$$a) F(x) = 19\ln|x| + k \rightarrow F'(x) = \frac{19}{x}$$

$$b) F(x) = \frac{e^{2x}}{2} + k \rightarrow F'(x) = e^{2x}$$

$$c) F(x) = \frac{\ln|x|}{19} + k \rightarrow F'(x) = \frac{1}{19x}$$

$$d) F(x) = \frac{e^{5x+55}}{5} + k \rightarrow F'(x) = e^{5x+55}$$

$$e) F(x) = 2\ln|2x+3| + k \rightarrow$$

$$\rightarrow F'(x) = \frac{4}{2x+3}$$

$$f) F(x) = \frac{2^x}{\ln 2} + k \rightarrow F'(x) = 2^x$$

- 50** Troba mentalment una primitiva de cadascuna d'aquestes funcions trigonomètriques i comprova la resposta.

$$a) f(x) = \cos x$$

$$b) f(x) = \cos(3x)$$

$$c) f(x) = \cos(x+3)$$

$$d) f(x) = 3 \sin x$$

$$e) f(x) = \sin(x-\pi)$$

$$f) f(x) = 3 \sin(x-\pi)$$

$$a) F(x) = \sin x + k \rightarrow F'(x) = \cos x$$

$$b) F(x) = \frac{\sin(3x)}{3} + k \rightarrow F'(x) = \cos(3x)$$

$$c) F(x) = \sin(x+3) + k \rightarrow$$

$$\rightarrow F'(x) = \cos(x+3)$$

$$d) F(x) = -3\cos x + k \rightarrow F'(x) = 3\sin x$$

$$e) F(x) = -\cos(x-\pi) + k \rightarrow$$

$$\rightarrow F'(x) = \sin(x-\pi)$$

$$f) F(x) = -3\cos(x-\pi) + k \rightarrow$$

$$\rightarrow F'(x) = 3\sin(x-\pi)$$

- 51** **INVESTIGA.** Respon les preguntes següents.

a) Per què una primitiva d'una funció polinòmica $f(x)$ és una altra funció polinòmica $F(x)$?

b) Si $f(x)$ és de grau n , quin és el grau de $F(x)$?

a) Perquè la derivada d'una funció polinòmica sempre és una altra funció polinòmica.

b) El grau de $F(x)$ es $n+1$, si considerem que $f(x)$ és un polinomi i que n sigui diferent de -1 .

Integrals de funcions elementals



ACTIVITATS FLAIX

- 52** Calcula les integrals polinòmiques següents.

$$a) \int (2x-3) dx$$

$$b) \int (3x^2+4x-2) dx$$

$$c) \int \left(\frac{3}{4}x^3 - 3x^2 + 6x - 1 \right) dx$$

$$d) \int \left(\frac{x^3}{3} - \frac{x^2}{2} + x - 5 \right) dx$$

$$e) \int (x+3)^2 dx$$

$$f) \int (1-2x)^2 dx$$

$$g) \int (x-1)^3 dx$$

$$h) \int (x^2 - 3x + 2) dx$$

$$a) \int (2x - 3) dx = x^2 - 3x + k$$

$$b) \int (3x^2 + 4x - 2) dx = x^3 + 2x^2 - 2x + k$$

$$c) \int \left(\frac{3}{4}x^3 - 3x^2 + 6x - 1 \right) dx = \frac{3}{16}x^4 - x^3 + 3x^2 - x + k$$

$$d) \int \left(\frac{x^3}{3} - \frac{x^2}{2} + x - 5 \right) dx = \frac{x^4}{12} - \frac{x^3}{6} + \frac{x^2}{2} - 5x + k$$

$$e) \int (x + 3)^2 dx = \frac{(x + 3)^3}{3} + k$$

$$f) \int (1 - 2x)^2 dx = \frac{-1}{2} \int (-2) \cdot (1 - 2x)^2 dx = \frac{-1}{2} \cdot \frac{(1 - 2x)^3}{3} + k = -\frac{1}{6}(1 - 2x)^3 + k$$

$$g) \int (x - 1)^3 dx = \frac{(x - 1)^4}{4} + k$$

$$h) \int (x^2 - 3x + 2) dx = \frac{x^3}{3} - 3\frac{x^2}{2} + 2x + k$$

53 Troba les integrals d'aquestes funcions amb radicals.

$$a) \int 2\sqrt{x} dx \quad c) \int \left(\frac{1}{\sqrt{x}} - x \right) dx$$

$$b) \int 2\sqrt[3]{x} dx \quad d) \int (\sqrt[4]{x} + \sqrt[3]{x}) dx$$

$$a) \int 2\sqrt{x} dx = \frac{4}{3}\sqrt{x^3} + k$$

$$b) \int 2\sqrt[3]{x} dx = 2\sqrt[3]{3} \int x^{\frac{1}{3}} dx = \frac{4\sqrt[3]{3}}{3}x^{\frac{4}{3}} + k = \frac{4}{3}\sqrt[3]{3x^4} + k$$

$$c) \int \left(\frac{1}{\sqrt{x}} - x \right) dx = 2\sqrt{x} - \frac{x^2}{2} + k$$

$$d) \int (\sqrt[4]{x} + \sqrt[3]{x}) dx = \frac{4}{5}\sqrt[4]{x^5} + \frac{3}{4}\sqrt[3]{x^4} + k$$

54 Determina aquestes integrals de funcions racionals.

$$a) \int \frac{4}{x-2} dx$$

$$b) \int \left(\frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3} \right) dx$$

$$c) \int \left(\frac{2}{x+3} - \frac{1}{x-4} \right) dx$$

$$d) \int \frac{1}{(x+4)^2} dx$$

$$e) \int \left(\frac{1}{(x-1)^2} + \frac{7}{x+3} \right) dx$$

$$f) \int \left(\frac{2}{(x+3)^3} - \frac{5}{(x-3)^2} \right) dx$$

$$a) \int \frac{4}{x-2} dx = 4\ln|x+2| + k$$

$$b) \int \left(\frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3} \right) dx = \ln|x| - \frac{2}{x} - \frac{3}{2x^2} + k$$

$$c) \int \left(\frac{2}{x+3} - \frac{1}{x-4} \right) dx = 2\ln|x+3| - \ln|x-4| + k$$

$$d) \int \frac{1}{(x+4)^2} dx = -\frac{1}{x+4} + k$$

$$e) \int \left(\frac{1}{(x-1)^2} + \frac{7}{x+3} \right) dx = \frac{-1}{x-1} + 7\ln|x+3| + k$$

$$f) \int \left(\frac{2}{(x+3)^3} + \frac{5}{(x-3)^2} \right) dx =$$

$$= \frac{-1}{(x+3)^2} - \frac{5}{(x-3)} + k$$

55 REpte. Demuestra que

$$\int 2^x \cdot 3^x \cdot \dots \cdot n^x dx = \frac{(n!)^x}{\ln(n!)} + k.$$

$$2^x \cdot 3^x \cdot \dots \cdot n^x = (2 \cdot 3 \cdot \dots \cdot n)^x = (n!)^x$$

$$\int (n!)^x dx = \frac{(n!)^x}{\ln(n!)} + k$$

56 Troba les integrals de les funcions exponencials següents.

a) $\int e^{x-2} dx$ d) $\int \left(xe^{x^2} + \frac{4}{3}x^3\right) dx$

b) $\int (e^x + 1) dx$ e) $\int e^{2x+3} dx$

c) $\int (e^{2x} + 2^x) dx$ f) $\int (2e^x - 3x^2) dx$

a) $\int e^{x-2} dx = e^{x-2} + k$

b) $\int (e^x + 1) dx = e^x + x + k$

c) $\int (e^{2x} + 2^x) dx = \frac{1}{2}e^{2x} + \frac{2^x}{\ln 2} + k$

d) $\int \left(xe^{x^2} + \frac{4}{3}x^3\right) dx = \frac{1}{2}e^{x^2} + \frac{1}{3}x^4 + k$

e) $\int e^{2x+3} dx = \frac{1}{2}e^{2x+3} + k$

f) $\int (2e^x - 3x^2) dx = 2e^x - x^3 + k$

57 Calcula les integrals d'aquestes funcions.

a) $\int \cos(2x) dx$ f) $\int \frac{7}{\sin^2(3x)} dx$

b) $\int 4 \sin(x + \pi) dx$ g) $\int \frac{5}{\cos^2 \frac{x}{3}} dx$

c) $\int 3 \cos\left(\frac{\pi}{2} - \frac{x}{3}\right) dx$ h) $\int \frac{3}{\sqrt{1-x^2}} dx$

d) $\int 5 \sin(2x - \pi) dx$ i) $\int \frac{3}{x^2 + 1} dx$

e) $\int 3 \sec^2\left(\frac{1}{5}x\right) dx$ j) $\int \frac{1}{(3x)^2 + 1} dx$

a) $\int \cos(2x) dx = \frac{1}{2} \sin(2x) + k$

b) $\int 4 \sin(x + \pi) dx =$
 $= -4 \cos(x + \pi) + k$

c) $\int 3 \cos\left(\frac{\pi}{2} - \frac{x}{3}\right) dx =$
 $= -9 \sin\left(\frac{\pi}{2} - \frac{x}{3}\right) + k$

d) $\int 5 \sin(2x - \pi) dx =$
 $= -\frac{5}{2} \cos(2x - \pi) + k$

e) $\int 3 \sec^2\left(\frac{1}{5}x\right) dx = 15 \int \frac{\frac{1}{5} dx}{\cos^2\left(\frac{1}{5}x\right)} =$
 $= 15 \operatorname{tg}\left(\frac{1}{5}x\right) + k$

f) $\int \frac{7}{\sin^2(3x)} dx = \frac{7}{3} \int \frac{3}{\sin^2(3x)} dx =$
 $= -\frac{7}{3} \operatorname{cotg}(3x) + k$

g) $\int \frac{5}{\cos^2 \frac{x}{3}} dx = 15 \int \frac{\frac{1}{3}}{\cos^2 \frac{x}{3}} dx =$
 $= 15 \operatorname{tg} \frac{x}{3} + k$

h) $\int \frac{3}{\sqrt{1-x^2}} dx = 3 \arcsin x + k$

i) $\int \frac{3}{x^2 + 1} dx = 3 \operatorname{arctg} x + k$

j) $\int \frac{1}{(3x)^2 + 1} dx =$
 $= \frac{1}{3} \operatorname{arctg}(3x) + k$

58 INVESTIGA. Comprova, amb alguns exemples, si les afirmacions següents són certes.

a) La integral d'un producte de funcions no coincideix amb el producte d'integrals.

- b) La integral d'un quocient de funcions no coincideix amb el quocient d'integrals.

Resposta oberta. Per exemple:

$$a) \int x^2 dx = \frac{x^3}{3} + k$$

$$\int x dx = \frac{x^2}{2}$$

$$\int x dx \cdot \int x dx = \frac{x^4}{4} + k$$

Aleshores l'afirmació és certa.

$$b) \int 1 dx = \int \frac{x}{x} dx = x + k$$

$$\int x dx = \frac{x^2}{2}$$

$$\frac{\int x dx}{\int x dx} = \frac{\frac{x^2}{2}}{\frac{x^2}{2}} + k = 1 + k$$

Aleshores l'afirmació és certa.

59 Resol aquestes integrals.

$$a) \int \frac{2x}{x^2 + 1} dx$$

$$b) \int \frac{8x - 3}{4x^2 - 3x + 1} dx$$

$$c) \int \frac{\sin x}{\cos x} dx$$

$$d) \int \cotg x dx$$

$$e) \int (2x + 1) \sin(x^2 + x) dx$$

$$f) \int 6x \cos(3x^2 - 5) dx$$

$$g) \int 6x e^{3x^2} dx$$

$$h) \int (3x^2 + 1) e^{x^3 + x} dx$$

$$i) \int x e^{7x^2} dx$$

$$j) \int \frac{-2}{4 + x^2} dx$$

$$k) \int x e^{x^2} dx$$

$$l) \int \left(\frac{x + \ln x}{x} \right) dx$$

$$a) \int \frac{2x}{x^2 + 1} dx = \ln|x^2 + 1| + k$$

$$b) \int \frac{8x - 3}{4x^2 - 3x + 1} dx = \ln|4x^2 - 3x + 1| + k$$

$$c) \int \frac{\sin x}{\cos x} dx = -\ln|\cos x| + k$$

$$d) \int \cotg x dx = \int \frac{\cos x}{\sin x} dx = \ln|\sin x| + k$$

$$e) \int (2x + 1) \sin(x^2 + x + 5) dx = -\cos(x^2 + x + 5) + k$$

$$f) \int 6x \cos(3x^2 - 5) dx = \sin(3x^2 - 5) + k$$

$$g) \int 6x e^{3x^2} dx = e^{3x^2} + k$$

$$h) \int (3x^2 + 1) e^{x^3 + x} dx = e^{x^3 + x} + k$$

$$i) \int x e^{7x^2} dx = \frac{e^{7x^2}}{14} + k$$

$$j) \int \frac{-2}{4 + x^2} dx = \frac{-1}{2} \int \frac{1}{1 + \left(\frac{x}{2}\right)^2} dx = -\arctg \frac{x}{2} + k$$

$$t = x^2 \rightarrow dt = 2x dx$$

$$k) \int x^3 e^{x^2} dx = \frac{1}{2} \int t e^t dt = \frac{1}{2} e^{x^2} (x^2 - 1) + k$$

$$l) \int \frac{x + \ln x}{x} dx = \int dx + \int \frac{\ln x}{x} dx = x + \int \frac{\ln x}{x} dx = x + \frac{\ln^2 x}{2} + k$$

60 REpte. Resol la integral

$$\int \cos x \sqrt{1 + \tg^2 x} dx.$$

$$\sqrt{1 + \operatorname{tg}^2 x} = \sqrt{\sec^2 x} = \sec x = \frac{1}{\cos x}$$

$$\int \cos x \cdot \frac{1}{\cos x} dx = \int dx = x + k$$

61 Fes aquestes integrals.

•••

a) $\int \frac{x}{x^2 - 3} dx$

e) $\int \frac{2 \cos x}{3 + \sin x} dx$

b) $\int \frac{x^3}{\sqrt{x^4 + 3}} dx$

f) $\int \frac{x}{x^2 + 2^4} dx$

c) $\int \frac{\sin x}{1 - \cos x} dx$

g) $\int \frac{x}{\sqrt{1 + 3x^2}} dx$

d) $\int \frac{3 + x}{\sqrt{1 - x^2}} dx$

h) $\int \frac{x - 1}{x^2 + 1} dx$

a) $\int \frac{x}{x^2 - 3} dx = \frac{1}{2} \int \frac{2x}{x^2 - 3} dx =$
 $= \frac{1}{2} \ln |x^2 - 3| + k$

$$t = x^4 + 3 \rightarrow dt = 4x^3 dx \rightarrow x^3 dx = \frac{1}{4} dt$$

b) $\int \frac{x^3}{\sqrt{x^4 + 3}} dx = \frac{1}{4} \int \frac{1}{\sqrt{t}} dt =$
 $= \frac{1}{2} \sqrt{t} + k = \frac{1}{2} \sqrt{x^4 + 3} + k$

c) $\int \frac{\sin x}{1 - \cos x} dx = \ln |1 - \cos x| + k$

d) $\int \frac{3 + x}{\sqrt{1 - x^2}} dx = \int \frac{3}{\sqrt{1 - x^2}} dx +$

$$t = 1 - x^2 \rightarrow dt = -2x dx \rightarrow x dx = -\frac{1}{2} dt$$

$$+ \int \frac{x}{\sqrt{1 - x^2}} dx =$$

$$= 3 \arcsin x - \frac{1}{2} \int \frac{1}{\sqrt{t}} dt =$$

$$= 3 \arcsin x - \sqrt{1 - x^2} + k$$

e) $\int \frac{2 \cos x}{3 + \sin x} dx = 2 \ln |3 + \sin x| + k$

$$t = x^2 + 2^4 \rightarrow dt = 2x dx$$

f) $\int \frac{x}{x^2 + 2^4} dx = \frac{1}{2} \int \frac{dt}{t} =$
 $= \frac{1}{2} \ln |x^2 + 2^4| + k$

$$t = 1 + 3x^2 \rightarrow dt = 6x dx \rightarrow x dx = \frac{1}{6} dt$$

g) $\int \frac{x}{\sqrt{1 + 3x^2}} dx = \frac{1}{6} \int \frac{1}{\sqrt{t}} dt =$
 $= \frac{2}{6} \sqrt{t} + k = \frac{1}{3} \sqrt{1 + 3x^2} + k$

h) $\int \frac{x - 1}{x^2 + 1} dx =$
 $= \int \frac{x}{x^2 + 1} dx + \int \frac{-1}{x^2 + 1} dx =$
 $= \frac{1}{2} \ln |x^2 + 1| - \arctg x + k$

62 **INVESTIGA.** Resol aquestes integrals.

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$$\int (\operatorname{tg}^2 x + n) dx$$

$$\int (\operatorname{tg}^2 x - n) dx$$

$$\int \operatorname{tg}^2 x dx + \int n dx = \operatorname{tg} x - x + nx + k =$$

 $= \operatorname{tg} x + x(n - 1)$

$$\int \operatorname{tg}^2 x dx - \int n dx = \operatorname{tg} x - x - nx + k =$$

 $= \operatorname{tg} x - x(n + 1)$

63 La gràfica d'una funció passa pel punt $A(0, 1)$. Si la seva derivada és $f'(x) = (x + 1)(x^2 + 2x + 6)$, quina és l'expressió algebraica de la funció?

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$$F(x) = \int (x + 1)(x^2 + 2x + 6) dx =$$

$$= \int (x^3 + 3x^2 + 8x + 6) dx =$$

$$= \frac{x^4}{4} + x^3 + 4x^2 + 6x + k$$

$$F(0) = 1 \rightarrow k = 1 \rightarrow$$

$$\rightarrow F(x) = \frac{x^4}{4} + x^3 + 4x^2 + 6x + 1$$

64 Una funció $y = f(x)$, amb $x > -1$, té per

•••

derivada: $y' = \frac{a}{1 + x}$, en què

a és una constant. Determina la funció si, a més, sabem que $f(0) = 1$ i $f(1) = -1$.

$$f(x) = \int \frac{a}{1+x} dx = a \ln|1+x| + k$$

$$f(0) = 1 \rightarrow a \ln 1 + k \rightarrow k = 1$$

$$f(1) = -1 \rightarrow a \ln 2 + 1 = -1 \rightarrow a = \frac{-2}{\ln 2}$$

$$\text{La funció és } f(x) = \frac{-2}{\ln 2} \ln|1+x| + 1.$$

65 Troba la primitiva de la funció

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$f(x) = \frac{-x}{1-x^2}$ la gràfica de la qual passa pel punt de coordenades $A(\sqrt{2}, 3)$.

$$F(x) = \int \frac{-x}{1-x^2} dx = \frac{1}{2} \int \frac{-2x}{1-x^2} dx = \frac{1}{2} \ln|1-x^2| + k$$

$$F(\sqrt{2}) = 3 \rightarrow \frac{1}{2} \ln|1-2| + k = 3 \rightarrow k = 3$$

La funció primitiva és:

$$F(x) = \frac{1}{2} \ln|1-x^2| + 3$$

66 Calcula una funció real f sabent que:

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- Té un punt d'inflexió en l'origen de coordenades.
- La recta tangent en l'origen de coordenades és $y = 5x$.
- Verifica la condició $f'''(x) = 24x - 6$.

Sabem que f passa per l'origen de coordenades, pel que $f(0) = 0$.

A més, aquest és un punt d'inflexió, alhora $f''(0) = 0$.

Com que el pendent de la recta tangent en $(0, 0)$ és 5, podem concloure que $f'(0) = 5$.

Finalment, tenim:

$$f'''(x) = 24x - 6$$

$$f''(x) = \int (24x - 6) dx = 12x^2 - 6x + k \rightarrow$$

$$\rightarrow f''(0) = k = 0 \rightarrow f''(x) = 12x^2 - 6x$$

$$f'(x) = \int (12x^2 - 6x) dx =$$

$$= 4x^3 - 3x^2 + k \rightarrow f'(0) = 5 \rightarrow$$

$$\rightarrow k = 5 \rightarrow f'(x) = 4x^3 - 3x^2 + 5$$

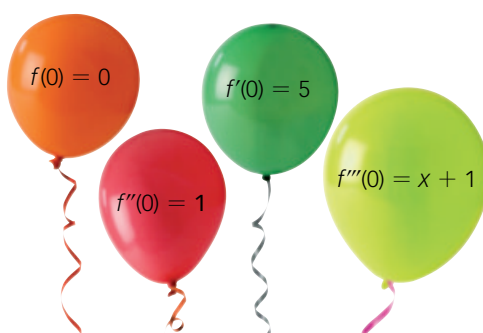
$$f(x) = \int (4x^3 - 3x^2 + 5) dx =$$

$$= x^4 - x^3 + 5x + k \rightarrow f(0) = 0 \rightarrow$$

$$\rightarrow k = 0 \rightarrow f(x) = x^4 - x^3 + 5x$$

67 Troba l'expressió algebraica d'una funció que compleix les condicions següents.

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$$f''(x) = \int (x+1) dx = \frac{x^2}{2} + x + k \rightarrow$$

$$\rightarrow f''(0) = 1 \rightarrow k = 1 \rightarrow$$

$$\rightarrow f''(x) = \frac{x^2}{2} + x + 1$$

$$f'(x) = \int \left(\frac{x^2}{2} + x + 1 \right) dx =$$

$$= \frac{x^3}{6} + \frac{x^2}{2} + x + k \rightarrow f'(0) = 5 \rightarrow$$

$$\rightarrow k = 5 \rightarrow f'(x) = \frac{x^3}{6} + \frac{x^2}{2} + x + 5$$

$$f(x) = \int \left(\frac{x^3}{6} + \frac{x^2}{2} + x + 5 \right) dx =$$

$$= \frac{x^4}{24} + \frac{x^3}{6} + \frac{x^2}{2} + 5x + k \rightarrow f(0) = 0 \rightarrow$$

$$\rightarrow k = 0 \rightarrow f(x) = \frac{x^4}{24} + \frac{x^3}{6} + \frac{x^2}{2} + 5x$$

68 Calcula la primitiva de la funció següent sabent que passa pel punt $P(0, 2)$.

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$$f(x) = \frac{1 - \sin x}{2x + 2\cos x}$$

$$F(x) = \int \frac{1 - \sin x}{2x + 2\cos x} dx =$$

$$= \frac{1}{2} \int \frac{1 - \sin x}{x + \cos x} dx = \frac{1}{2} \ln|x + \cos x| + k$$

$$F(0) = 2 \rightarrow 0 + k = 2 \rightarrow k = 2 \rightarrow$$

$$\rightarrow F(x) = \frac{1}{2} \ln|x + \cos x| + 2$$

- 69 **INVESTIGA.** Troba l'equació de dues funcions les gràfiques de les quals passen pel punt $A(0, 1)$ i la derivada segona és $f''(x) = \sin x$.

$$f'(x) = \int \sin x dx = -\cos x + k$$

$$f(x) = \int (-\cos x + k) dx = -\sin x + kx + l$$

$$f(0) = 1 \rightarrow l = 1 \rightarrow f(x) = -\sin x + kx + 1$$

Resposta oberta. Per exemple:

$$f(x) = -\sin x + x + 1$$

$$f(x) = -\sin x - 2x + 1$$

- 70 La derivada segona d'una funció és $f''(x) = 6x + 2$. Troba'n l'expressió algebraica sabent que té un mínim relatiu en el punt $A(1, 3)$.

$$f'(x) = \int (6x + 2) dx = 3x^2 + 2x + k$$

f té un mínim relatiu en $A(1, 3)$;

per tant, $f(1) = 3$ i $f'(1) = 0$.

$$f'(1) = 0 \rightarrow 3 + 2 + k = 0 \rightarrow k = -5 \rightarrow$$

$$\rightarrow f'(x) = 3x^2 + 2x - 5$$

$$f(x) = \int (3x^2 + 2x - 5) dx =$$

$$= x^3 + x^2 - 5x + k \rightarrow f(1) = 3 \rightarrow$$

$$\rightarrow 1 + 1 - 5 + k = 3 \rightarrow k = 6 \rightarrow$$

$$\rightarrow f(x) = x^3 + x^2 - 5x + 6$$

- 71 Donada la funció:

$$f(x) = \frac{3x^2 + \cos x + 2e^{2x}}{x^3 + \sin x + e^{2x}}$$

- a) Determina la primitiva que passa pel punt $A(0, -5)$.
b) Alguna primitiva d'aquesta funció passa per l'origen de coordenades? I pel punt $B(0, 1)$?

$$a) F(x) = \int \frac{3x^2 + \cos x + 2 \cdot e^{2x}}{x^3 + \sin x + e^{2x}} dx =$$

$$= \ln|x^3 + \sin x + e^{2x}| + k \rightarrow$$

$$\rightarrow F(0) = -5 \rightarrow k = -5 \rightarrow$$

$$\rightarrow F(x) = \ln|x^3 + \sin x + e^{2x}| - 5$$

$$b) F(0) = k \rightarrow k = 0 \rightarrow$$

$\rightarrow F(x) = \ln|x^3 + \sin x + e^{2x}|$ passa per l'origen de coordenades.

$$F(0) = k \rightarrow k = 1 \rightarrow$$

$\rightarrow F(x) = \ln|x^3 + \sin x + e^{2x}| + 1$ passa pel punt $B(0, 1)$.

- 72 **REpte.** D'una funció derivable se sap que passa pel punt $(-1, -4)$ i que la seva derivada és:

$$f'(x) = \begin{cases} 2 - x & \text{si } x \leq 1 \\ \frac{1}{x} & \text{si } x > 1 \end{cases}$$

- a) Troba l'expressió de $f(x)$.
b) Determina l'equació de la recta tangent a $f(x)$ en el punt d'abscissa $x = 2$.

$$a) f(-1) = -4$$

$$f(x) = \begin{cases} f_1(x) & \text{si } x \leq 1 \\ f_2(x) & \text{si } x > 1 \end{cases}$$

$$f_1(x) = \int (2 - x) dx = 2x - \frac{x^2}{2} + k \rightarrow$$

$$\rightarrow f_1(-1) = -4 \rightarrow$$

$$\rightarrow -2 - \frac{1}{2} + k = -4 \rightarrow k = -\frac{3}{2} \rightarrow$$

$$\rightarrow f_1(x) = 2x - \frac{x^2}{2} - \frac{3}{2} \text{ si } x \leq 1$$

$$f_2(x) = \int \frac{1}{x} dx = \ln|x| + k \text{ si } x > 1$$

Com que f és derivable, llavors és contínua; per tant:

$$f_1(1) = f_2(1) \rightarrow 2 - \frac{1}{2} - \frac{3}{2} = 0 = k \rightarrow$$

$$\rightarrow f_2(x) = \ln|x|$$

Aleshores:

$$f(x) = \begin{cases} 2x - \frac{x^2}{2} - \frac{3}{2} & \text{si } x \leq 1 \\ \ln|x| & \text{si } x > 1 \end{cases}$$

- b) El pendent de la recta tangent en el punt $x = 2$ és $f'(2) = \frac{1}{2}$. A més, $f(2) = \ln 2$. Per tant, la recta és:
 $y = \frac{1}{2}x + n \rightarrow \ln 2 = \frac{1}{2} \cdot 2 + n \rightarrow$
 $\rightarrow n = \ln 2 - 1 \rightarrow y = \frac{1}{2}x + \ln 2 - 1$

2. Aplica els mètodes bàsics per calcular primitives de funcions

Integració per parts

73 Calcula per parts aquestes integrals.

- a) $\int x^3 \ln x \, dx$ g) $\int x^2 \sin(2x) \, dx$
 b) $\int \ln(2x+1) \, dx$ h) $\int (2x+3) e^{2x} \, dx$
 c) $\int e^{-x} \sin(2x) \, dx$ i) $\int \frac{x}{e^x} \, dx$
 d) $\int \arctg x \, dx$ j) $\int (x^2 - 5) \cos x \, dx$
 e) $\int \frac{\ln x}{x} \, dx$ k) $\int (2x^2 + x - 2) e^{3x} \, dx$
 f) $\int x \sin(2x) \, dx$ l) $\int (2 + e^{2x}) \cos(x+1) \, dx$

$$u = \ln x \rightarrow du = \frac{1}{x} dx$$

$$\begin{aligned} \text{a) } \int x^3 \ln x \, dx &= \frac{x^4 \ln x}{4} - \int \frac{x^3}{4} \, dx = \\ &= \frac{x^4 \ln x}{4} - \frac{x^4}{16} + k = \frac{x^4}{4} \left(\ln x - \frac{1}{4} \right) + k \end{aligned}$$

$$u = \ln(2x+1) \rightarrow du = \frac{2}{2x+1} dx$$

$$\begin{aligned} \text{b) } \int \ln(2x+1) \, dx &= \\ &= x \ln|2x+1| - \int \frac{2x}{2x+1} \, dx = \\ &= x \ln|2x+1| - \int \left(1 - \frac{1}{2x+1} \right) \, dx = \\ &= x \ln|2x+1| - x + \frac{\ln|2x+1|}{2} + k = \\ &= \left(x + \frac{1}{2} \right) \ln|2x+1| - x + k \end{aligned}$$

$$\begin{aligned} u &= e^{-x} \rightarrow du = -e^{-x} dx \\ dv &= \sin(2x) dx \rightarrow v = \frac{-\cos(2x)}{2} \end{aligned}$$

$$\text{c) } I = \int e^{-x} \sin(2x) \, dx =$$

$$\begin{aligned} u &= e^{-x} \rightarrow du = -e^{-x} dx \\ dv &= \cos(2x) dx \rightarrow v = \frac{\sin(2x)}{2} \end{aligned}$$

$$\begin{aligned} &= -e^{-x} \frac{\cos(2x)}{2} - \frac{1}{2} \int e^{-x} \cos(2x) \, dx = \\ &= -e^{-x} \frac{\cos(2x)}{2} - \\ &- \frac{1}{2} \left(e^{-x} \frac{\sin(2x)}{2} + \frac{1}{2} \int e^{-x} \sin(2x) \, dx \right) = \\ &= -e^{-x} \frac{\cos(2x)}{2} - e^{-x} \frac{\sin(2x)}{4} - \\ &- \frac{1}{4} \int e^{-x} \sin(2x) \, dx = \\ &= -e^{-x} \frac{\cos(2x)}{2} - e^{-x} \frac{\sin(2x)}{4} - \frac{1}{4} I \\ I &= -e^{-x} \frac{\cos(2x)}{2} - e^{-x} \frac{\sin(2x)}{4} - \frac{1}{4} I \rightarrow \\ &\rightarrow \frac{5}{4} I = -e^{-x} \left(\frac{\cos(2x)}{2} + \frac{\sin(2x)}{4} \right) \rightarrow \\ &\rightarrow I = -\frac{e^{-x}}{5} (2\cos(2x) + \sin(2x)) + k \end{aligned}$$

$$u = \arctg x \rightarrow du = \frac{1}{x^2+1} dx$$

$$\begin{aligned} \text{d) } \int \arctg x \, dx &= \\ &= x \cdot \arctg x - \int \frac{x}{x^2+1} \, dx = \\ &= x \cdot \arctg x - \frac{\ln|x^2+1|}{2} + k \end{aligned}$$

$$u = \ln x \rightarrow du = \frac{1}{x} dx$$

$$\begin{aligned} \text{e) } I &= \int \frac{\ln x}{x} \, dx = \ln^2 x - \int \frac{\ln x}{x} \, dx = \\ &= \ln^2 x - I \rightarrow I = \ln^2 x - I \rightarrow \\ &\rightarrow I = \int \frac{\ln x}{x} \, dx = \frac{1}{2} \ln^2 x + k \end{aligned}$$

$$u = x \rightarrow du = dx$$

$$\begin{aligned} \text{f) } \int x \sin(2x) \, dx &= \\ &= \frac{-x \cos(2x)}{2} + \int \frac{\cos(2x)}{2} \, dx = \\ &= -\frac{x \cos(2x)}{2} + \frac{\sin(2x)}{4} + k \end{aligned}$$

$$\begin{aligned} u &= x^2 \rightarrow du = 2x dx \\ dv &= \sin(2x) dx \rightarrow v = \frac{-\cos(2x)}{2} \end{aligned}$$

g) $\int x^2 \sin(2x) dx =$

$$\begin{aligned} u &= x^2 \rightarrow du = 2x dx \\ dv &= \sin(2x) dx \rightarrow v = \frac{-\cos(2x)}{2} \end{aligned}$$

$$\begin{aligned} &= \frac{-x^2 \cos(2x)}{2} + \int x \cos(2x) dx = \\ &= \frac{-x^2 \cos(2x)}{2} + \frac{x \sin(2x)}{2} - \\ &\quad - \int \frac{\sin(2x)}{2} dx = \\ &= \frac{-x^2 \cos(2x)}{2} + \frac{x \sin(2x)}{2} + \frac{\cos(2x)}{4} + k \end{aligned}$$

$$u = 2x + 3 \rightarrow du = 2x dx$$

h) $\int (2x + 3)e^{2x} dx = \frac{(2x + 3)e^{2x}}{2} -$
 $-\int e^{2x} dx = \frac{(2x + 3)e^{2x}}{2} - \frac{e^{2x}}{2} + k =$
 $= (x + 1)e^{2x} + k$

$$u = x \rightarrow du = dx$$

i) $\int \frac{x}{e^x} dx = -\frac{x}{e^x} + \int e^{-x} dx =$
 $= -\frac{x}{e^x} - \frac{1}{e^x} + k = \frac{-x - 1}{e^x} + k$

$$\begin{aligned} u &= x^2 - 5 \rightarrow du = 2x dx \\ dv &= \cos x dx \rightarrow v = \sin x \end{aligned}$$

j) $\int (x^2 - 5) \cos x dx = (x^2 - 5) \sin x -$

$$\begin{aligned} u &= x \rightarrow du = dx \\ dv &= \sin x dx \rightarrow v = -\cos x \end{aligned}$$

$$\begin{aligned} &-2 \int x \sin x dx = (x^2 - 5) \sin x + \\ &-2x \cos x - 2 \int \cos x dx = \\ &= (x^2 - 5) \sin x + 2x \cos x - 2 \sin x + \\ &+ k = (x^2 - 7) \sin x + 2x \cos x + k \end{aligned}$$

$$\begin{aligned} u &= 2x^2 + x - 2 \rightarrow du = (4x + 1) dx \\ dv &= e^{3x} dx \rightarrow v = \frac{e^{3x}}{3} \end{aligned}$$

k) $\int (2x^2 + x - 2)e^{3x} dx =$

$$= \frac{(2x^2 + x - 2)e^{3x}}{3} -$$

$$\begin{aligned} u &= 4x + 1 \rightarrow du = 4 dx \\ dv &= e^{3x} dx \rightarrow v = \frac{e^{3x}}{3} \end{aligned}$$

$$\begin{aligned} &- \int \frac{(4x + 1)e^{3x}}{3} dx = \\ &= \frac{(2x^2 + x - 2)e^{3x}}{3} - \\ &- \frac{1}{3} \left(\frac{(4x + 1)e^{3x}}{3} - \int \frac{4e^{3x}}{3} dx \right) = \\ &= \frac{(2x^2 + x - 2)e^{3x}}{3} - \frac{(4x + 1)e^{3x}}{9} + \\ &+ \frac{4}{27} e^{3x} + k = \frac{1}{27} e^{3x} (18x^2 - \\ &- 3x - 17) + k \end{aligned}$$

l) $\int (2 + e^{2x}) \cos(x + 1) dx =$
 $= \int 2 \cos(x + 1) dx + \int e^{2x} \cos(x + 1) dx =$

$$\begin{aligned} u &= e^{2x} \rightarrow du = 2e^{2x} dx \\ dv &= \cos(x + 1) dx \rightarrow v = \sin(x + 1) \end{aligned}$$

$$= 2 \sin(x + 1) + e^{2x} \sin(x + 1) -$$

$$\begin{aligned} u &= e^{2x} \rightarrow du = 2e^{2x} dx \\ dv &= \sin(x + 1) dx \rightarrow v = -\cos(x + 1) \end{aligned}$$

$$\begin{aligned} &- \int 2e^{2x} \sin(x + 1) dx = \\ &= 2 \sin(x + 1) + e^{2x} \sin(x + 1) + \\ &+ 2e^{2x} \cos(x + 1) - \\ &- 2 \int 2e^{2x} \cos(x + 1) dx = \\ &= 2 \sin(x + 1) + \\ &+ \frac{e^{2x} \sin(x + 1) + 2e^{2x} \cos(x + 1)}{5} + k \end{aligned}$$

74 INVENTA. Escriu una funció la integral de la qual es pugui resoldre pel mètode d'integració per parts i resol-la.

Resposta oberta. Per exemple: $\int x e^x dx$

$$u = x \rightarrow du = dx$$

$$dv = e^x dx \rightarrow v = e^x$$

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + k$$

75 INVESTIGA. Fes servir el mètode d'integració per parts per calcular les integrals de les funcions següents.

- a) $f(x) = \ln x^2$
 b) $f(x) = \ln x^3$
 c) $f(x) = \ln x^4$
 d) $f(x) = \ln x^5$
 e) $f(x) = \ln x^n$, en què n és un nombre enter.

$$u = \ln x^2 \rightarrow du = \frac{2x}{x^2} dx$$

$$\begin{aligned} \text{a) } \int \ln x^2 dx &= x \ln x^2 - \int x \cdot \frac{2x}{x^2} dx = \\ &= x \ln x^2 - \int 2 dx = x \ln x^2 - 2x + k \end{aligned}$$

$$u = \ln x^3 \rightarrow du = \frac{3x^2}{x^3} dx$$

$$\begin{aligned} \text{b) } \int \ln x^3 dx &= x \ln x^3 - \int x \cdot \frac{3x^2}{x^3} dx = \\ &= x \ln x^3 - \int 3 dx = x \ln x^3 - 3x + k \end{aligned}$$

$$u = \ln x^4 \rightarrow du = \frac{4x^3}{x^4} dx$$

$$\begin{aligned} \text{c) } \int \ln x^4 dx &= x \ln x^4 - \int x \cdot \frac{4x^3}{x^4} dx = \\ &= x \ln x^4 - \int 4 dx = x \ln x^4 - 4x + k \end{aligned}$$

$$u = \ln x^5 \rightarrow du = \frac{5x^4}{x^5} dx$$

$$\begin{aligned} \text{d) } \int \ln x^5 dx &= x \ln x^5 - \int x \cdot \frac{5x^4}{x^5} dx = \\ &= x \ln x^5 - \int 5 dx = x \ln x^5 - 5x + k \end{aligned}$$

$$u = \ln x^n \rightarrow du = \frac{nx^{n-1}}{x^n} dx$$

$$\begin{aligned} \text{e) } \int \ln x^n dx &= \\ &= x \ln x^n - \int x \cdot \frac{nx^{n-1}}{x^n} dx = \\ &= x \ln x^n - \int n dx = x \ln x^n - nx + k \end{aligned}$$

76 Calcula les integrals següents.

- a) $\int x^2 e^{3x} dx$ e) $\int x 2^x dx$
 b) $\int \ln \left(\frac{x+1}{x-1} \right)^x dx$ f) $\int x \ln^2 x dx$
 c) $\int \frac{\ln x}{x^3} dx$ g) $\int \arcsin x dx$
 d) $\int x \operatorname{arctg}(x+1) dx$ h) $\int \frac{x \arcsin x}{\sqrt{1-x^2}} dx$

$$\begin{aligned} u &= x^2 \rightarrow du = 2x dx \\ dv &= e^{3x} dx \rightarrow v = \frac{1}{3} e^{3x} \end{aligned}$$

$$\text{a) } \int x^2 \cdot e^{3x} dx = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \int x e^{3x} dx =$$

$$\begin{aligned} u &= x \rightarrow du = dx \\ dv &= e^{3x} dx \rightarrow v = \frac{1}{3} e^{3x} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \left[\frac{1}{3} x e^{3x} - \frac{1}{3} \int e^{3x} dx \right] = \\ &= \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{9} \cdot \frac{1}{3} e^{3x} + k = \\ &= \frac{e^{3x}}{3} \left(x^2 - \frac{2}{3} x + \frac{2}{9} \right) + k \end{aligned}$$

$$\text{b) } \int \ln \left(\frac{x+1}{x-1} \right)^x dx = \int x \ln \left(\frac{x+1}{x-1} \right) dx =$$

$$\begin{aligned} u &= x \rightarrow du = dx \\ dv &= \ln(x+1) dx \rightarrow v = \frac{1}{x+1} \end{aligned}$$

$$= \int x \ln(x+1) dx - \int x \ln(x-1) dx =$$

$$\begin{aligned} u &= x \rightarrow du = dx \\ dv &= \ln(x-1) dx \rightarrow v = \frac{1}{x-1} \end{aligned}$$

$$\begin{aligned} &= \frac{x}{x+1} - \int \frac{1}{x+1} dx - \frac{x}{x-1} + \\ &+ \int \frac{1}{x-1} dx = \frac{x}{x+1} - \ln|x+1| + \\ &+ 1 - \frac{x}{x-1} + \ln|x-1| + k \end{aligned}$$

$$u = \ln x \rightarrow du = \frac{1}{x} dx$$

$$\text{c) } \int \frac{\ln x}{x^3} dx = -\frac{\ln x}{2x^2} + \frac{1}{2} \int \frac{1}{x^3} dx =$$

$$= -\frac{\ln x}{2x^2} - \frac{1}{4x^2} + k$$

d) $\int x \operatorname{arctg}(x+1) dx = \frac{1}{2} x^2 \operatorname{arctg}(x+1) -$

$$u = \operatorname{arctg}(x+1) \rightarrow du = \frac{1}{1+(x+1)^2} dx$$

$$- \int \frac{x^2}{2(x^2+2x+2)} dx =$$

$$= \frac{1}{2} x^2 \operatorname{arctg}(x+1) -$$

$$- \frac{1}{2} \int \left(1 - \frac{2(x+1)}{x^2+2x+2} \right) dx =$$

$$= \frac{1}{2} x^2 \operatorname{arctg}(x+1) - \frac{1}{2} \int dx +$$

$$+ \frac{1}{2} \int \frac{2(x+1)}{x^2+2x+2} dx =$$

$$= \frac{1}{2} x^2 \operatorname{arctg}(x+1) - \frac{1}{2} x +$$

$$+ \frac{1}{2} \ln |x^2+2x+2| + k$$

$$u = x \rightarrow du = dx$$

$$dv = 2^x dx \rightarrow v = \frac{2^x}{\ln 2}$$

e) $\int x \cdot 2^x dx = x \cdot \frac{2^x}{\ln 2} - \int \frac{2^x}{\ln 2} dx =$

$$= x \cdot \frac{2^x}{\ln 2} - \frac{1}{\ln 2} \int 2^x dx =$$

$$= x \cdot \frac{2^x}{\ln 2} - \frac{1}{\ln 2} \cdot \frac{1}{\ln 2} \int \ln 2 \cdot 2^x dx =$$

$$= \frac{2^x x}{\ln 2} - \frac{2^x}{\ln^2 2} + k$$

$$u = \ln^2 x \rightarrow du = \frac{2 \ln x}{x} dx$$

$$dv = x dx \rightarrow v = \frac{x^2}{2}$$

f) $\int x \cdot \ln^2 x dx = \ln^2 x \cdot \frac{x^2}{2} - \int x \ln x dx =$

$$u = \ln x \rightarrow du = \frac{1}{x} dx$$

$$dv = x dx \rightarrow v = \frac{x^2}{2}$$

$$= \ln^2 x \cdot \frac{x^2}{2} - \left(\frac{x^2}{2} \ln x - \frac{1}{2} \int x dx \right) =$$

$$= \ln^2 x \cdot \frac{x^2}{2} - \frac{x^2}{2} \ln x - \frac{1}{4} x^2 + k =$$

$$= \frac{1}{2} x^2 \left(\ln^2 x - \ln x - \frac{1}{2} \right) + k$$

$$u = \arcsin x \rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$$

g) $\int \arcsin x dx =$

$$= x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx =$$

$$= x \arcsin x + \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx =$$

$$= x \arcsin x + \sqrt{1-x^2} + k$$

$$t = \arcsin x \rightarrow dt = \frac{1}{\sqrt{1-x^2}} dx$$

h) $\int \frac{x \cdot \arcsin x}{\sqrt{1-x^2}} dx = \int t \sin t dt =$

$$u = t \rightarrow du = dt$$

$$dv = \sin t dt \rightarrow v = -\cos t$$

$$= -t \cos t + \int \cos t dt =$$

$$= -t \cos t + \sin t + k =$$

$$= -\cos(\arcsin x) \cdot \arcsin x + x + k =$$

$$= -\arcsin x \cdot \sqrt{1-x^2} + x + k$$

77 Fes els càlculs que calgui per obtenir el valor d'aquestes integrals.

a) $\int \sqrt[3]{x} \ln x dx$ d) $\int \ln(1+x^2) dx$

b) $\int (x^2+3x) e^{-x+7} dx$ e) $\int x^2 \operatorname{arctg} x dx$

c) $\int e^x \cos(3x) dx$ f) $\int \frac{\sqrt{x}+1}{\sqrt[3]{x}} \ln x dx$

$$u = \ln x \rightarrow du = \frac{1}{x} dx$$

a) $\int \sqrt[3]{x} \ln x dx =$

$$= \frac{3}{4} \ln x \sqrt[3]{x^4} - \frac{3}{4} \int \sqrt[3]{x} dx =$$

$$= \frac{3}{4} \ln x \sqrt[3]{x^4} - \frac{3}{4} \cdot \frac{3}{4} \sqrt[3]{x^4} + k =$$

$$= \frac{3}{4} \ln x \sqrt[3]{x^4} - \frac{9}{16} \sqrt[3]{x^4} + k$$

$$\begin{aligned} u &= x^2 + 3x \rightarrow du = (2x + 3)dx \\ dv &= e^{-x+7} dx \rightarrow v = -e^{-x+7} \end{aligned}$$

$$b) \int (x^2 + 3x)e^{-x+7} dx =$$

$$= -(x^2 + 3x)e^{-x+7} +$$

$$\begin{aligned} u &= 2x + 3 \rightarrow du = 2dx \\ dv &= e^{-x+7} dx \rightarrow v = -e^{-x+7} \end{aligned}$$

$$+ \int (2x + 3)e^{-x+7} dx =$$

$$= -(x^2 + 3x)e^{-x+7} - (2x + 3)e^{-x+7} - 2e^{-x+7} + k$$

$$\begin{aligned} u &= e^x \rightarrow du = e^x dx \\ dv &= \cos(3x) dx \rightarrow v = \frac{1}{3} \sin(3x) \end{aligned}$$

$$c) \int e^x \cos(3x) dx =$$

$$\begin{aligned} u &= e^x \rightarrow du = e^x dx \\ dv &= \sin(3x) dx \rightarrow v = \frac{1}{3} \cos(3x) \end{aligned}$$

$$= \frac{1}{3} e^x \sin(3x) - \frac{1}{3} \int e^x \sin(3x) dx =$$

$$= \frac{1}{3} e^x \sin(3x) -$$

$$- \frac{1}{3} \left(-\frac{1}{3} \cos(3x) + \frac{1}{3} \int e^x \cos(3x) dx \right) =$$

$$= \frac{1}{3} e^x \sin(3x) + \frac{1}{9} \cos(3x) -$$

$$- \frac{1}{9} \int e^x \cos(3x) dx$$

$$I = \frac{1}{3} e^x \sin(3x) + \frac{1}{9} \cos(3x) - \frac{1}{9} I \rightarrow$$

$$\rightarrow I = \frac{9}{10} \left(\frac{1}{3} e^x \sin(3x) + \frac{1}{9} \cos(3x) \right) \rightarrow$$

$$\rightarrow I = \frac{1}{10} (3e^x \sin(3x) + \cos(3x))$$

$$\begin{aligned} u &= \ln(1 + x^2) \rightarrow du = \frac{2x}{1 + x^2} dx \\ dv &= dx \rightarrow v = x \end{aligned}$$

$$d) \int \ln(1 + x^2) dx =$$

$$= x \ln(1 + x^2) - \int \frac{2x^2}{x^2 + 1} dx =$$

$$= x \ln(1 + x^2) - \int \left(2 - \frac{2}{x^2 + 1} \right) dx =$$

$$= x \ln(1 + x^2) - 2x - 2 \arctg x + k$$

$$\begin{aligned} u &= \arctg x \rightarrow du = \frac{1}{x^2 + 1} dx \\ dv &= x^2 dx \rightarrow v = \frac{x^3}{3} \end{aligned}$$

$$e) \int x^2 \cdot \arctg x dx =$$

$$= \frac{x^3}{3} \arctg x - \int \frac{x^3}{3(x^2 + 1)} dx =$$

$$= \frac{x^3}{3} \arctg x - \frac{1}{3} \int \left(x - \frac{x}{x^2 + 1} \right) dx =$$

$$= \frac{x^3}{3} \arctg x - \frac{1}{6} x^2 + \frac{1}{6} \ln |x^2 + 1| + k$$

$$f) \int \frac{\sqrt{x} + 1}{\sqrt[3]{x}} \ln x dx =$$

$$\begin{aligned} u &= \ln x \rightarrow du = \frac{1}{x} dx \\ dv &= (x^{1/6} + x^{-1/2}) dx \rightarrow v = \frac{6}{7} x^{7/6} + \frac{3}{2} x^{2/3} \end{aligned}$$

$$= \int (x^{1/6} + x^{-1/3}) \ln x dx =$$

$$= \ln x \left(\frac{6}{7} x^{7/6} + \frac{3}{2} x^{2/3} \right) -$$

$$- \int \left(\frac{6}{7} x^{1/6} + \frac{3}{2} x^{-1/3} \right) dx =$$

$$= \left(\frac{6}{7} x^{7/6} + \frac{3}{2} x^{2/3} \right) \ln x - \frac{36}{49} x^{7/6} -$$

$$- \frac{9}{4} x^{2/3} + k$$

78 Aplica el mètode d'integració per parts i resol la integral següent.

$$\int x \sin(2x + 1) \cos(2x - 1) dx$$

$$\int x \sin(2x + 1) \cos(2x - 1) dx =$$

$$= \frac{1}{2} \int x (\sin(2) + \sin(4x)) dx =$$

$$\begin{aligned} u &= x \rightarrow du = dx \\ dv &= \sin(4x) dx \rightarrow v = -\frac{1}{4}\cos(4x) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2}\sin(2) \int x dx + \frac{1}{2} \int x \sin(4x) dx = \\ &= \frac{1}{4}\sin(2)x^2 + \frac{1}{2} \left(-\frac{1}{4}x\cos(4x) + \right. \\ &\quad \left. + \frac{1}{4} \int \cos 4x dx \right) = \frac{1}{4}x^2\sin 2 - \\ &\quad - \frac{1}{8}x\cos(4x) + \frac{1}{32}\sin(4x) + k \end{aligned}$$

- 79 **INVESTIGA.** La derivada de la funció $f(x)$ és $f'(x) = x^2 \sin x$. Troba l'expressió algebraica de $f(x)$ sabent que talla l'eix d'ordenades en el valor $y = 1$.

$$\begin{aligned} u &= x^2 \rightarrow du = 2x dx \\ dv &= \sin x dx \rightarrow v = -\cos x \end{aligned}$$

$$f(x) = \int x^2 \cdot \sin x dx =$$

$$\begin{aligned} u &= x \rightarrow du = dx \\ dv &= \cos x dx \rightarrow v = \sin x \end{aligned}$$

$$\begin{aligned} &= -x^2 \cos x + 2 \int x \cos x dx = \\ &= -x^2 \cos x + 2 \left(x \sin x - \int \sin x dx \right) = \\ &= -x^2 \cos x + 2x \sin x + 2 \cos x + k \\ f(0) &= 1 \rightarrow 2 \cos(0) + k = 1 \rightarrow \\ &\rightarrow k = 1 - 2 = -1 \\ f(x) &= -x^2 \cos x + 2x \sin x + 2 \cos x - 1 \end{aligned}$$

- 80 Troba la funció f que verifica les condicions següents.

• $f(e) = \frac{e}{2}$ • $f'(1) = 0$ • $f''(x) = 2x \ln x$

$$\begin{aligned} u &= \ln x \rightarrow du = \frac{1}{x} dx \\ dv &= x dx \rightarrow v = \frac{x^2}{2} \end{aligned}$$

$$\begin{aligned} f'(x) &= \int f''(x) dx = \int 2x \cdot \ln x dx = \\ &= 2 \left(\frac{1}{2} x^2 \ln x - \int \frac{1}{2} x dx \right) = \\ &= x^2 \ln x - \frac{1}{2} x^2 + k \end{aligned}$$

$$f'(1) = 0 \rightarrow \ln 1 - \frac{1}{2} + k = 0 \rightarrow k = \frac{1}{2}$$

$$f'(x) = x^2 \ln x - \frac{1}{2} x^2 + \frac{1}{2}$$

$$f(x) = \int f'(x) = \frac{1}{6} x^3 + \frac{1}{2} x =$$

$$\begin{aligned} &= \int \left(x^2 \ln x - \frac{1}{2} x^2 + \frac{1}{2} \right) dx = \\ &= \int x^2 \ln x dx - \frac{1}{2} \int x^2 dx + \frac{1}{2} \int dx = \end{aligned}$$

$$\begin{aligned} u &= \ln x \rightarrow du = \frac{1}{x} dx \\ dv &= x^2 dx \rightarrow v = \frac{x^3}{3} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 dx - \frac{1}{6} x^3 + \frac{1}{2} x = \\ &= \frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 - \frac{1}{6} x^3 + \frac{1}{2} x + k = \\ &= \left(\frac{1}{3} \ln x - \frac{5}{18} \right) x^3 + \frac{1}{2} x + k \end{aligned}$$

$$f(e) = \frac{e}{2} \rightarrow$$

$$\rightarrow \left(\frac{1}{3} - \frac{5}{18} \right) e^3 + \frac{1}{2} e + k = \frac{e}{2} \rightarrow$$

$$\rightarrow k = -\frac{1}{18} e^3$$

$$f(x) = \left(\frac{1}{3} \ln x - \frac{5}{18} \right) x^3 + \frac{1}{2} x - \frac{1}{18} e^3$$

- 81 Troba la primitiva de la funció

$$f(x) = \frac{3}{1+x^2} \text{ que passa pel punt}$$

$$P\left(1, \frac{\pi}{2}\right).$$

$$\begin{aligned} F(x) &= \int f(x) dx = \int \frac{3}{1+x^2} dx = \\ &= 3 \arctg(x) + k \end{aligned}$$

$$F(1) = \frac{\pi}{2} \rightarrow 3 \arctg(1) + k = \frac{\pi}{2} \rightarrow$$

$$\rightarrow k = \frac{\pi}{2} - \frac{3\pi}{4} \rightarrow k = -\frac{\pi}{4} \rightarrow$$

$$\rightarrow F(x) = 3 \arctg(x) - \frac{\pi}{4}$$

- 82 Calcula el valor de a perquè
una primitiva de la funció
 $f(x) = ax^2 + x \cdot \cos x + 1$ passi pel punt
 $P(\pi, \pi)$.

$$F(x) = \int f(x) dx =$$

$$= \int (ax^2 + x \cdot \cos x + 1) dx =$$

$$\begin{array}{l} u = x \rightarrow du = dx \\ dv = \cos x dx \rightarrow v = \sin x \end{array}$$

$$= a \int x^2 dx + \int x \cos x dx + \int dx =$$

$$= \frac{a}{3} x^3 + x \sin x - \int \sin x dx + x =$$

$$= \frac{a}{3} x^3 + x \sin x + \cos x + x + k$$

Si $k = 0$:

$$F(\pi) = \pi \rightarrow \frac{a}{3} \pi^3 - 1 + \pi = \pi \rightarrow a = \frac{3}{\pi^3}$$

- 83 En cada un dels casos següents,
troba una funció $f(x)$ que compleixi
les condicions:

a) $f'(x) = x^2 \sin x, f(0) = -1$

b) $f'(x) = x \ln x, f(1) = \frac{1}{2}$

$$\begin{array}{l} u = x^2 \rightarrow du = 2x dx \\ dv = \sin x dx \rightarrow v = -\cos x \end{array}$$

a) $f(x) = \int f'(x) dx = \int x^2 \sin x dx =$

$$\begin{array}{l} u = x \rightarrow du = dx \\ dv = \cos x dx \rightarrow v = \sin x \end{array}$$

$$= -x^2 \cos x + 2 \int x \cos x dx =$$

$$= -x^2 \cos x + 2x \sin x + 2 \cos x + k =$$

$$= (2 - x^2) \cos x + 2x \sin x + k$$

$$f(0) = -1 \rightarrow 2 + k = -1 \rightarrow k = -3 \rightarrow$$

$$\rightarrow f(x) = (2 - x^2) \cos x + 2x \sin x - 3$$

$$\begin{array}{l} u = \ln x \rightarrow du = \frac{1}{x} dx \\ dv = x dx \rightarrow v = \frac{x^2}{2} \end{array}$$

b) $f(x) = \int f'(x) dx = \int x \ln x dx =$

$$= \frac{x^2 \ln x}{2} - \int \frac{x}{2} dx = \frac{x^2 \ln x}{2} - \frac{x^2}{4} + k$$

$$f(1) = \frac{1}{2} \rightarrow -\frac{1}{4} + k = \frac{1}{2} \rightarrow$$

$$\rightarrow k = \frac{3}{4} \rightarrow f(x) = \frac{x^2 \ln x}{2} - \frac{x^2}{4} + \frac{3}{4}$$

- 84 **REpte.** Troba l'expressió de la família
de funcions tal que el pendent de
les rectes tangents a les gràfiques
en qualsevol punt ve donat per
 $m = (2x + 3) e^{2x}$. Determina, d'entre
totes, la que passa pel punt $(0, 1)$.

El pendent de la recta tangent és el valor
de la derivada de la funció.

$$F(x) = \int (2x + 3) e^{2x} dx =$$

$$= \frac{(2x + 3) e^{2x}}{2} - \int e^{2x} dx =$$

$$= \frac{(2x + 3) e^{2x}}{2} - \frac{e^{2x}}{2} + k =$$

$$= \frac{(2x + 3) e^{2x} - e^{2x}}{2} + k =$$

$$= (x + 1) e^{2x} + k \rightarrow F(0) = 1 \rightarrow$$

$$\rightarrow 1 + k = 1 \rightarrow k = 0 \rightarrow F(x) = (x + 1) e^{2x}$$

Integrals de funcions racionals

- 85 Calcula les integrals següents.

a) $\int \frac{3}{x^2 - 3x + 2} dx$

b) $\int \frac{2 - x}{x^2 + 3x} dx$

c) $\int \frac{x - 3}{x^2 - 4} dx$

a) $\int \frac{3}{x^2 - 3x + 2} dx =$

$$= \int \left(\frac{-3}{x - 1} + \frac{3}{x - 2} \right) dx =$$

$$= \int \frac{-3}{x - 1} dx + \int \frac{3}{x - 2} dx =$$

$$= -3 \ln |x - 1| + 3 \ln |x - 2| + k$$

$$\begin{aligned} \text{b) } \int \frac{2-x}{x^2+3x} dx &= \int \left(\frac{\frac{2}{3}}{x} + \frac{\frac{-5}{3}}{x+3} \right) dx = \\ &= \frac{2}{3} \int \frac{1}{x} dx + \frac{-5}{3} \int \frac{1}{x+3} dx = \\ &= \frac{2}{3} \ln|x| - \frac{5}{3} \ln|x+3| + k \end{aligned}$$

$$\begin{aligned} \text{c) } \int \frac{x-3}{x^2-4} dx &= \int \left(\frac{\frac{5}{4}}{x+2} + \frac{\frac{-1}{4}}{x-2} \right) dx = \\ &= \frac{5}{4} \int \frac{1}{x+2} dx - \frac{1}{4} \int \frac{1}{x-2} dx = \\ &= \frac{5}{4} \ln|x+2| - \frac{1}{4} \ln|x-2| + k \end{aligned}$$

86 Integra les funcions racionals següents.

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$$\text{a) } f(x) = \frac{3}{x^2-2x+1}$$

$$\text{b) } g(x) = \frac{x+2}{x^2-2x+1}$$

$$\text{c) } h(x) = \frac{x}{x^2-2x+1}$$

$$\text{d) } i(x) = \frac{x^2}{x^2-2x+1}$$

$$\begin{aligned} \text{a) } \int f(x) dx &= \int \frac{3}{x^2-2x+1} dx = \\ &= 3 \int \frac{1}{(x-1)^2} dx = \frac{-3}{x-1} + k \end{aligned}$$

$$\begin{aligned} \text{b) } \int g(x) dx &= \int \frac{x+2}{x^2-2x+1} dx = \\ &= \int \left(\frac{1}{(x-1)} + \frac{3}{(x-1)^2} \right) dx = \\ &= \int \frac{1}{(x-1)} dx + \int \frac{3}{(x-1)^2} dx = \\ &= \ln|x-1| - \frac{3}{x-1} + k \end{aligned}$$

$$\begin{aligned} \text{c) } \int h(x) dx &= \int \frac{x}{x^2-2x+1} dx = \\ &= \int \left(\frac{1}{(x-1)} + \frac{1}{(x-1)^2} \right) dx = \\ &= \int \frac{1}{(x-1)} dx + \int \frac{1}{(x-1)^2} dx = \end{aligned}$$

$$= \ln|x-1| - \frac{1}{x-1} + k$$

$$\begin{aligned} \text{d) } \int i(x) dx &= \int \frac{x^2}{x^2-2x+1} dx = \\ &= \int \left(1 + \frac{2x-1}{x^2-2x+1} \right) dx = \\ &= \int dx + \int \frac{2x-1}{x^2-2x+1} dx = \\ &= x + \int \left(\frac{2}{x-1} + \frac{1}{(x-1)^2} \right) dx = \\ &= x + 2\ln|x-1| - \frac{1}{x-1} + k \end{aligned}$$

87 Calcula aquestes integrals de funcions racionals.

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$$\text{a) } \int \frac{2}{x^3+x^2} dx \quad \text{c) } \int \frac{x+2}{x^3-x^2-x+1} dx$$

$$\text{b) } \int \frac{1+x^2}{3x^2-x^3} dx \quad \text{d) } \int \frac{1+x^3}{x^3-2x^2} dx$$

$$\begin{aligned} \text{a) } \int \frac{2}{x^3+x^2} dx &= \\ &= \int \left(-\frac{2}{x} + \frac{2}{x^2} + \frac{2}{x+1} \right) dx = \\ &= -2\ln|x| - \frac{2}{x} + 2\ln|x+1| + k \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{1+x^2}{3x^2-x^3} dx &= \\ &= \int \left(\frac{1}{\frac{9}{x}} + \frac{1}{\frac{3}{x^2}} + \frac{\frac{10}{9}}{3-x} \right) dx = \\ &= \frac{1}{9} \ln|x| - \frac{1}{3x} - \frac{10}{9} \ln|3-x| + k \end{aligned}$$

$$\begin{aligned} \text{c) } \int \frac{x+2}{x^3-x^2-x+1} dx &= \\ &= \int \left(\frac{-\frac{1}{4}}{x-1} + \frac{\frac{3}{2}}{(x-1)^2} + \frac{\frac{1}{4}}{x+1} \right) dx = \\ &= -\frac{1}{4} \ln|x-1| - \frac{3}{2} \cdot \frac{1}{x-1} + \\ &\quad + \frac{1}{4} \ln|x+1| + k \end{aligned}$$

$$\begin{aligned}
 \text{d) } \int \frac{1+x^3}{x^3-2x^2} dx &= \\
 &= \int \left(1 + \frac{2x^2+1}{x^3-2x^2} \right) dx = \\
 &= \int dx + \int \left(\frac{-1}{x} + \frac{-1}{x^2} + \frac{9}{x-2} \right) dx = \\
 &= x - \frac{1}{4} \ln|x| + \frac{1}{2x} + \frac{9}{4} \ln|x-2| + k
 \end{aligned}$$

88 Fes aquestes integrals indefinides.

$$\begin{aligned}
 \text{a) } \int \frac{2}{x^3+x} dx & \\
 \text{b) } \int \frac{x+2}{x^3+x^2+x+1} dx & \\
 \text{c) } \int \frac{x+3}{4x^3-4x^2+x-1} dx & \\
 \text{d) } \int \frac{1}{(x-2)^2(x^2+2)} dx & \\
 \text{e) } \int \frac{3x^2+1}{x^4-1} dx & \\
 \text{f) } \int \frac{x^2+x+1}{x^3-x^2-x+1} dx & \\
 \text{a) } \int \frac{2}{x^3+x} dx &= \int \left(\frac{2}{x} - \frac{2x}{x^2+1} \right) dx = \\
 &= 2 \ln|x| - \ln|x^2+1| + k \\
 \text{b) } \int \frac{x+2}{x^3+x^2+x+1} dx &= \\
 &= \int \left(\frac{3}{2(x^2+1)} - \frac{x}{2(x^2+1)} + \right. \\
 &\quad \left. + \frac{1}{2(x+1)} \right) dx = \frac{3}{2} \arctg x - \\
 &\quad - \frac{1}{4} \ln|x^2+1| + \frac{1}{2} \ln|x+1| + k \\
 \text{c) } \int \frac{x+3}{4x^3-4x^2+x-1} dx &= \\
 &= \int \left(\frac{-16x-11}{5(4x^2+1)} + \frac{4}{5(x-1)} \right) dx = \\
 &= \int \left(\frac{-16x}{5(4x^2+1)} - \frac{11}{5(4x^2+1)} + \right. \\
 &\quad \left. + \frac{4}{5(x-1)} \right) dx = -\frac{2}{5} \ln|4x^2+1| -
 \end{aligned}$$

$$\begin{aligned}
 &\quad - \frac{11}{10} \arctg(2x) + \frac{4}{5} \ln|x-1| + k \\
 \text{d) } \int \frac{1}{(x-2)^2(x^2+2)} dx &= \\
 &= \int \left(\frac{2x}{18(x^2+2)} + \frac{1}{18(x^2+2)} - \right. \\
 &\quad \left. - \frac{1}{9(x-2)} + \frac{1}{6(x-2)^2} \right) dx = \\
 &= \frac{1}{18} \ln|x^2+2| - \frac{1}{6(x-2)} - \\
 &\quad - \frac{1}{9} \ln|x-2| + \frac{\sqrt{2}}{36} \arctg\left(\frac{x}{\sqrt{2}}\right) + k \\
 \text{e) } \int \frac{3x^2+1}{x^4-1} dx &= \\
 &= \int \left(\frac{1}{x^2+1} - \frac{1}{x+1} + \frac{1}{x-1} \right) dx = \\
 &= \ln|1-x| - \ln|x+1| + \arctg x + k \\
 \text{f) } \int \frac{x^2+x+1}{x^3-x^2-x+1} dx &= \\
 &= \int \left(\frac{3}{4(x-1)} + \frac{3}{2(x-1)^2} + \right. \\
 &\quad \left. + \frac{1}{4(x+1)} \right) dx = \frac{3}{4} \ln|x-1| - \\
 &\quad - \frac{3}{2(x-1)} + \frac{1}{4} \ln|x+1| + k
 \end{aligned}$$

89 Calcula les integrals següents.

$$\begin{aligned}
 \text{a) } \int \frac{4x^3+2x-1}{2x+1} dx &\quad \text{c) } \int \frac{x^3}{x^2+4x-5} dx \\
 \text{b) } \int \frac{-x^2+x-1}{3-x} dx &\quad \text{d) } \int \frac{x^3-x+6}{x^2+5x+4} dx \\
 \text{a) } \int \frac{4x^3+2x-1}{2x+1} dx &= \\
 &= \int \left(2x^2 - x - \frac{5}{2(2x+1)} + \frac{3}{2} \right) dx = \\
 &= \frac{2}{3} x^3 - \frac{1}{2} x^2 - \frac{5}{4} \ln|2x+1| + \frac{3}{2} x + k \\
 \text{b) } \int \frac{-x^2+x-1}{3-x} dx &= \\
 &= \int \left(x+2 - \frac{7}{3-x} \right) dx =
 \end{aligned}$$

$$= \frac{1}{2}x^2 + 2x + 7\ln|3 - x| + k$$

$$\begin{aligned} \text{c) } \int \frac{x^3}{x^2 + 4x - 5} dx &= \\ &= \int \left(x + \frac{1}{6(x-1)} + \frac{125}{6(x+5)} - 4 \right) dx = \\ &= \frac{1}{2}(x-8)x + \frac{1}{6}\ln|1-x| + \\ &\quad + \frac{125}{6}\ln|x+5| + k \end{aligned}$$

$$\begin{aligned} \text{d) } \int \frac{x^3 - x + 6}{x^2 + 5x + 4} dx &= \\ &= \int \left(x + \frac{2}{x+1} + \frac{18}{x+4} - 5 \right) dx = \\ &= \frac{1}{2}x^2 + 2\ln|x+1| + \\ &\quad + 18\ln|x+4| - 5x + k \end{aligned}$$

90 ●●● Calcula les integrals de les funcions racionals següents.

$$\begin{array}{ll} \text{a) } \int \frac{x+3}{x-1} dx & \text{e) } \int \frac{x-2}{x^2-x} dx \\ \text{b) } \int \frac{2x+3}{2x+1} dx & \text{f) } \int \frac{x+2}{x^2-1} dx \\ \text{c) } \int \frac{3}{x^2+x-2} dx & \text{g) } \int \frac{1}{x^2+9} dx \\ \text{d) } \int \frac{5x-1}{x^2-1} dx & \text{h) } \int \frac{12}{x^3+4x^2+x-6} dx \end{array}$$

$$\begin{aligned} \text{a) } \int \frac{x+3}{x-1} dx &= \int \left(1 + \frac{4}{x-1} \right) dx = \\ &= x + 4\ln|x-1| + k \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{2x+3}{2x+1} dx &= \int \left(1 + \frac{2}{2x+1} \right) dx = \\ &= x + \ln|2x+1| + k \end{aligned}$$

$$\begin{aligned} \text{c) } \int \frac{3}{x^2+x-2} dx &= \\ &= \int \left(\frac{1}{x-1} - \frac{1}{x+2} \right) dx = \\ &= \ln|x-1| - \ln|x+2| + k \end{aligned}$$

$$\text{d) } \int \frac{5x-1}{x^2-1} dx =$$

$$\begin{aligned} &= \int \left(\frac{2}{x-1} + \frac{3}{x+1} \right) dx = \\ &= 2\ln|x-1| + 3\ln|x+1| + k \end{aligned}$$

$$\begin{aligned} \text{e) } \int \frac{x-2}{x^2-x} dx &= \int \left(\frac{-1}{x-1} + \frac{2}{x} \right) dx = \\ &= -\ln|x-1| + 2\ln|x| + k \end{aligned}$$

$$\begin{aligned} \text{f) } \int \frac{x+2}{x^2-1} dx &= \\ &= \int \left(\frac{3}{2(x-1)} - \frac{1}{2(x+1)} \right) dx = \\ &= \frac{3}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + k \end{aligned}$$

$$\text{g) } \int \frac{1}{x^2+9} dx = \frac{1}{3}\operatorname{arctg}\frac{x}{3} + k$$

$$\begin{aligned} \text{h) } \int \frac{12}{x^3+4x^2+x-6} dx &= \\ &= \int \left(\frac{1}{x-1} - \frac{4}{x+2} + \frac{3}{x+3} \right) dx = \\ &= \ln|x-1| - 4\ln|x+2| + \\ &\quad + 3\ln|x+3| + k \end{aligned}$$

91 ●●● Troba aquestes integrals de funcions racionals.

$$\begin{array}{ll} \text{a) } \int \frac{1}{x^2-5x+6} dx & \text{e) } \int \frac{x}{x^4+1} dx \\ \text{b) } \int \frac{x-2}{x^2+2x-3} dx & \text{f) } \int \frac{x^4-x^3-x-1}{x^3-x^2} dx \end{array}$$

$$\text{c) } \int \frac{x-4}{x^2+2x-3} dx \quad \text{g) } \int \frac{x^3}{(x+1)^4} dx$$

$$\text{d) } \int \frac{2x+8}{x^2-4} dx \quad \text{h) } \int \frac{2x+5}{(x+3)^3} dx$$

$$\begin{aligned} \text{a) } \int \frac{1}{x^2-5x+6} dx &= \\ &= \int \frac{1}{x-3} - \frac{1}{x-2} dx = \\ &= \ln|x-3| - \ln|x-2| + k \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{x-2}{x^2+2x-3} dx &= \\ &= \int \frac{1}{4} \left(\frac{5}{x+3} - \frac{1}{x-1} \right) dx = \\ &= \frac{5}{4}\ln|x+3| - \frac{1}{4}\ln|x-1| + k \end{aligned}$$

$$\begin{aligned} \text{c) } \int \frac{x-4}{x^2+2x-3} dx &= \\ &= \int \frac{1}{4} \left(\frac{7}{x+3} - \frac{3}{x-1} \right) dx = \\ &= \frac{7}{4} \ln|x+3| - \frac{3}{4} \ln|x-1| + k \end{aligned}$$

$$\begin{aligned} \text{d) } \int \frac{2x+8}{x^2-4} dx &= \\ &= \int \left(\frac{3}{x-2} - \frac{1}{x+2} \right) dx = \\ &= 3 \ln|x-2| - \ln|x+2| + k \end{aligned}$$

$$\text{e) } \int \frac{x}{x^4+1} dx = \frac{1}{2} \operatorname{arctg} x^2 + k$$

$$\begin{aligned} \text{f) } \int \frac{x^4-x^3-x+1}{x^3-x^2} dx &= \\ &= \int \left(x - \frac{1}{x^2} \right) dx = \frac{x^2}{2} + \frac{1}{x} + k \end{aligned}$$

$$\begin{aligned} \text{g) } \int \frac{x^3}{(x+1)^4} dx &= \\ &= \int \left(\frac{-1}{(x+1)^4} + \frac{3}{(x+1)^3} - \right. \\ &\quad \left. - \frac{3}{(x+1)^2} + \frac{1}{x+1} \right) dx = \\ &= \frac{1}{3(x+1)^3} - \frac{3}{2(x+1)^2} + \frac{3}{x+1} + \\ &\quad + \ln|x+1| + k \end{aligned}$$

$$\begin{aligned} \text{h) } \int \frac{2x+5}{(x+3)^3} dx &= \\ &= \int \left(\frac{-1}{(x+3)^3} + \frac{2}{(x+3)^2} \right) dx = \\ &= \frac{1}{2(x+3)^2} - \frac{2}{x+3} + k \end{aligned}$$

Integració per canvi de variable

92 Integra mitjançant un canvi de variable.

$$\text{a) } \int x^2 \cdot \sqrt{x^3+3} dx \quad \text{c) } \int \frac{2}{x \cdot \ln x} dx$$

$$\text{b) } \int x^3 \cdot e^{x^4+1} dx \quad \text{d) } \int \frac{\ln x^2}{x} dx$$

$$t = x^3 + 3 \rightarrow dt = 3x^2 dx \rightarrow \frac{1}{3x^2} dt = dx$$

$$\begin{aligned} \text{a) } \int x^2 \cdot \sqrt{x^3+3} dx &= \\ &= \frac{1}{3} \int \sqrt{t} dt = \frac{1}{3} \cdot \frac{2}{3} \sqrt{t^3} + k = \\ &= \frac{2}{9} \sqrt{t^3} + k = \frac{2}{9} \sqrt{(x^3+3)^3} + k \end{aligned}$$

$$t = x^4 + 1 \rightarrow dt = 4x^3 dx \rightarrow \frac{1}{4} dt = x^3 dx$$

$$\begin{aligned} \text{b) } \int x^3 \cdot e^{x^4+1} dx &= \int \frac{1}{4} e^t dt = \\ &= \frac{1}{4} e^t + k = \frac{1}{4} e^{x^4+1} + k \end{aligned}$$

$$t = \ln x \rightarrow dt = \frac{1}{x} dx$$

$$\begin{aligned} \text{c) } \int \frac{2}{x \cdot \ln x} dx &= 2 \int \frac{1}{t} dt = \\ &= 2 \ln|t| + k = 2 \ln|\ln|x|| + k \end{aligned}$$

$$\text{d) } \int \frac{\ln x^2}{x} dx = \int \frac{2 \ln x}{x} dx =$$

$$t = \ln x \rightarrow dt = \frac{1}{x} dx$$

$$\begin{aligned} &= 2 \int \frac{\ln x}{x} dx = 2 \int t dt = 2 \frac{t^2}{2} + k = \\ &= \ln^2 x + k \end{aligned}$$

93 Calcula aquestes integrals amb un canvi de variable.

$$\text{a) } \int \frac{e^{\sqrt{x}+1}}{\sqrt{x}} dx \quad \text{c) } \int \frac{e^{-\sqrt{x}+1}}{\sqrt{x}} dx$$

$$\text{b) } \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx \quad \text{d) } \int \frac{\sqrt{1+\sqrt{x}}}{\sqrt{x}} dx$$

Fem el canvi de variable a tots els apartats.

$$t = \sqrt{x} \rightarrow dt = \frac{1}{2\sqrt{x}} dx \rightarrow 2dt = \frac{1}{\sqrt{x}} dx$$

$$\begin{aligned} \text{a) } \int \frac{e^{\sqrt{x}+1}}{\sqrt{x}} dx &= 2 \int e^{t+1} dt = \\ &= 2e^{t+1} + k = 2e^{\sqrt{x}+1} + k \end{aligned}$$

$$\text{b) } \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = 2 \int \sin t dt =$$

$$= -2\cos t + k = -2\cos(\sqrt{x}) + k$$

$$c) \int \frac{e^{-\sqrt{x}+1}}{\sqrt{x}} dx = 2 \int e^{-t+1} dt =$$

$$= -2e^{-t+1} + k = -2e^{-\sqrt{x}+1} + k$$

$$d) \int \frac{\sqrt{1+\sqrt{x}}}{\sqrt{x}} dx = 2 \int \sqrt{1+t} dt =$$

$$= 2 \int (1+t)^{\frac{1}{2}} dt = \frac{4\sqrt{(1+t)^3}}{3} + k =$$

$$= \frac{4\sqrt{(1+\sqrt{x})^3}}{3} + k$$

94 Troba, mitjançant un canvi de variable, les integrals següents.

$$a) \int \cos x \sin^3 x dx \quad f) \int \cos^2 x \sin x dx$$

$$b) \int x \ln(1+x^2) dx \quad g) \int \sin x e^{\cos x} dx$$

$$c) \int \frac{\ln(2x)}{x} dx \quad h) \int \cos^2 x \sin^3 x dx$$

$$d) \int 2x \sin x^2 dx \quad i) \int \frac{\sin^3 x}{\cos x} dx$$

$$e) \int x \sqrt{x+1} dx \quad j) \int (x^2+1)e^{x^3+3x} dx$$

$$t = \sin x \rightarrow dt = \cos x dx$$

$$a) \int \cos x \sin^3 x dx = \int t^3 dt = \frac{t^4}{4} + k =$$

$$= \frac{\sin^4 x}{4} + k$$

$$t = 1+x^2 \rightarrow dt = 2x dx$$

$$b) \int x \ln(1+x^2) dx = \frac{1}{2} \int \ln t dt =$$

$$= \frac{1}{2} (t \ln t - t) + k =$$

$$= \frac{1}{2} ((1+x^2) \ln(1+x^2) - (1+x^2)) + k$$

$$t = \ln(2x) \rightarrow dt = \frac{1}{x} dx$$

$$c) \int \frac{\ln(2x)}{x} dx = \int t dt = \frac{t^2}{2} + k =$$

$$= \frac{\ln^2(2x)}{2} + k$$

$$t = x^2 \rightarrow dt = 2x dx$$

$$d) \int 2x \sin x^2 dx = \int \sin t dt =$$

$$= -\cos t + k = -\cos x^2 + k$$

$$t = \sqrt{x+1} \rightarrow t^2 = x+1 \rightarrow 2t dt = dx$$

$$e) \int x \sqrt{x+1} dx = \int 2(t^2-1)t^2 dt =$$

$$= \frac{2}{5} t^5 - \frac{2}{3} t^3 + k =$$

$$= \sqrt{x+1} \left(\frac{2(x+1)^2}{5} - \frac{2(x+1)}{3} \right) + k$$

$$t = \cos x \rightarrow dt = -\sin x dx$$

$$f) \int \cos^2 x \sin x dx = \int -t^2 dt =$$

$$= \frac{-t^3}{3} + k = \frac{-\cos^3 x}{3} + k$$

$$t = \cos x \rightarrow dt = -\sin x dx$$

$$g) \int \sin x e^{\cos x} dx = \int -e^t dt =$$

$$= -e^t + k = -e^{\cos x} + k$$

$$t = \cos x \rightarrow dt = -\sin x dx$$

$$h) \int \cos^2 x \sin^3 x dx = \int -t^2(1-t^2) dt =$$

$$= -\frac{t^3}{3} + \frac{t^5}{5} + k =$$

$$= -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + k$$

$$t = \cos x \rightarrow dt = -\sin x dx$$

$$i) \int \frac{\sin^3 x}{\cos x} dx = \int \frac{t^2-1}{t} dt =$$

$$= \frac{t^2}{2} - \ln |t| + k =$$

$$= \frac{\cos^2 x}{2} - \ln |\cos x| + k$$

$$t = x^3 + 3x \rightarrow dt = (3x^2 + 3) dx$$

$$j) \int (x^2+1)e^{x^3+3x} dx = \int \frac{e^t}{3} dt =$$

$$= \frac{e^t}{3} + k = \frac{e^{x^3+3x}}{3} + k$$

95 Resol fent un canvi de variable.

a) $\int \frac{2}{4+x^2} dx$ f) $\int \frac{x}{\sqrt{1-x^2}} dx$

b) $\int x(x+5)^{10} dx$ g) $\int \operatorname{tg} 2x dx$

c) $\int \frac{\operatorname{tg} \sqrt{x}}{\sqrt{x}} dx$ h) $\int \operatorname{cotg} \frac{x}{5} dx$

d) $\int x e^{3x^2} dx$ i) $\int \frac{x^4}{\sqrt{1-x^{10}}} dx$

e) $\int \frac{x}{\sqrt{1-x^4}} dx$ j) $\int e^x \sqrt{(e^x+1)^3} dx$

$t = \frac{x}{2} \rightarrow dt = \frac{1}{2} dx$

a) $\int \frac{2}{4+x^2} dx = \int \frac{1}{1+t^2} dt =$
 $= \operatorname{arctg} t + k = \operatorname{arctg} \frac{x}{2} + k$

$t = x+5 \rightarrow dt = dx$

b) $\int x(x+5)^{10} dx = \int (t-5)t^{10} dt =$
 $= \int (t^{11} - 5t^{10}) dt = \frac{t^{12}}{12} - \frac{5t^{11}}{11} + k =$
 $= \frac{(x+5)^{12}}{12} - \frac{5(x+5)^{11}}{11} + k$

$t = \sqrt{x} \rightarrow dt = \frac{1}{2\sqrt{x}} dx$

c) $\int \frac{\operatorname{tg} \sqrt{x}}{\sqrt{x}} dx = \int 2 \operatorname{tg} t dt =$
 $= -2 \ln |\cos t| + k =$
 $= -2 \ln |\cos \sqrt{x}| + k$

$t = 3x^2 \rightarrow dt = 6x dx$

d) $\int x e^{3x^2} dx = \int \frac{e^t}{6} dt = \frac{e^t}{6} + k =$
 $= \frac{e^{3x^2}}{6} + k$

$t = x^2 \rightarrow dt = 2x dx$

e) $\int \frac{x}{\sqrt{1-x^4}} dx = \frac{1}{2} \int \frac{1}{\sqrt{1-t^2}} dt =$
 $= \frac{1}{2} \operatorname{arcsint} t + k = \frac{1}{2} \operatorname{arcsin} x^2 + k$

$t = 1-x^2 \rightarrow dt = -2x dx$

f) $\int \frac{x}{\sqrt{1-x^2}} dx = \int \frac{-1}{2\sqrt{t}} dt =$
 $= -\sqrt{t} + k = -\sqrt{1-x^2} + k$

$t = \cos(2x) \rightarrow dt = -2 \sin(2x) dx$

g) $\int \operatorname{tg}(2x) dx = \int \frac{-1}{2t} dt =$
 $= \frac{-1}{2} \ln |t| + k = -\frac{1}{2} \ln |\cos(2x)| + k$

$t = \sin \frac{x}{5} \rightarrow dt = \frac{1}{5} \cos \frac{x}{5} dx$

h) $\int \operatorname{cotg} \frac{x}{5} dx = \int 5t dt = 5 \ln |t| + k =$
 $= 5 \ln \left| \sin \frac{x}{5} \right| + k$

$t = x^5 \rightarrow dt = 5x^4 dx$

i) $\int \frac{x^4}{\sqrt{1-x^{10}}} dx = \frac{1}{5} \int \frac{1}{\sqrt{1-t^2}} dt =$
 $= \frac{1}{5} \operatorname{arcsint} t + k = \frac{1}{5} \operatorname{arcsin} x^5 + k$

$t = e^x + 1 \rightarrow dt = e^x dx$

j) $\int e^x \sqrt{(e^x+1)^3} dx = \int \sqrt{t^3} dt =$
 $= \frac{2}{5} \sqrt{t^5} + k = \frac{2}{5} \sqrt{(e^x+1)^5} + k$

96 Calcula aquestes integrals.

a) $\int \cos^3 x \sin^5 x dx$

b) $\int \frac{\sin^3 x}{\cos^2 x} dx$

c) $\int \frac{\sin x \cos x}{\cos^2 x - 2 \sin^2 x} dx$

d) $\int \frac{1}{4 - 3 \cos^2 x + 5 \sin^2 x} dx$

e) $\int \frac{\sin^3 x}{\sqrt[3]{\cos x}} dx$

a) $\int \cos^3 x \sin^5 x dx =$



$$\begin{aligned}
 & \boxed{t = \sin x \rightarrow dt = \cos x dx} \\
 & = \int \cos x (1 - \sin^2 x) \sin^5 x dx = \\
 & = \int (1 - t^2) t^5 dt = \int t^5 dt - \int t^7 dt = \\
 & = \frac{1}{6} t^6 - \frac{1}{8} t^8 + k = \\
 & = \frac{1}{6} \sin^6 x - \frac{1}{8} \sin^8 x + k
 \end{aligned}$$

$$\begin{aligned}
 & \boxed{t = \cos x \rightarrow dt = -\sin x dx} \\
 \text{b)} \int \frac{\sin^3 x}{\cos^2 x} dx &= \int \frac{(1 - \cos^2 x) \sin x}{\cos^2 x} dx = \\
 &= \int \frac{t^2 - 1}{t^2} dt = \int dt - \int \frac{1}{t^2} dt = \\
 &= t + \frac{1}{t} + k = \cos x + \frac{1}{\cos x} + k
 \end{aligned}$$

$$\begin{aligned}
 & \boxed{t = \cos^2 x - 2\sin^2 x \rightarrow dt = -6\cos x \sin x dx} \\
 \text{c)} \int \frac{\sin x \cdot \cos x}{\cos^2 x - 2\sin^2 x} dx &= -\frac{1}{6} \int \frac{1}{t} dt = \\
 &= -\frac{1}{6} \ln |t| + k = \\
 &= -\frac{1}{6} \ln |\cos^2 x - 2\sin^2 x| + k \\
 \text{d)} \int \frac{1}{4 - 3\cos^2 x + 5\sin^2 x} dx &= \\
 &= \int \frac{\operatorname{cosec}^2 x}{(4 - 3\cos^2 x + 5\sin^2 x) \operatorname{cosec}^2 x} dx = \\
 &= \int \frac{\operatorname{cosec}^2 x}{4 \operatorname{cosec}^2 x - 3 \cot^2 x + 5} dx =
 \end{aligned}$$

$$\begin{aligned}
 & \boxed{t = \cot x \rightarrow dt = -\operatorname{cosec}^2 x dx} \\
 &= \int \frac{\operatorname{cosec}^2 x}{9 + \cot^2 x} dx = - \int \frac{1}{t^2 + 9} dt = \\
 &= \frac{1}{3} \operatorname{arctg} \frac{t}{3} + k = \\
 &= \frac{1}{3} \operatorname{arctg} \left(\frac{\cot x}{3} \right) + k
 \end{aligned}$$

$$\begin{aligned}
 \text{e)} \int \frac{\sin^3 x}{\sqrt[3]{\cos x}} dx &= \\
 & \boxed{t = \cos x \rightarrow dt = -\sin x dx} \\
 &= \int \frac{\sin x (1 - \cos^2 x)}{\sqrt[3]{\cos x}} dx =
 \end{aligned}$$

$$\begin{aligned}
 & \boxed{t = \sin x \rightarrow dt = \cos x dx} \\
 &= - \int \frac{1 - t^2}{\sqrt[3]{t}} dt = \\
 &= - \int t^{-\frac{1}{3}} dt + \int \frac{t^5}{t^{\frac{1}{3}}} dt = \\
 &= -\frac{3}{2} \sqrt[3]{\cos^2 x} + \frac{3\sqrt[3]{\cos^8 x}}{8} + k
 \end{aligned}$$

97 Integra fent servir un canvi de variable.

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$$\begin{aligned}
 \text{a)} \int \frac{1 + \sqrt{x+1}}{1 + \sqrt[3]{x+1}} dx & \quad \text{d)} \int \frac{x + \sqrt{x}}{\sqrt{x} + \sqrt[4]{x}} dx \\
 \text{b)} \int \frac{\sqrt{x}}{1 + \sqrt[3]{x}} dx & \quad \text{e)} \int \frac{1}{\sqrt{x}(1 + \sqrt[4]{x})} dx \\
 \text{c)} \int \frac{1 + x + \sqrt{x+1}}{(x+1)\sqrt[3]{x+1}} dx & \quad \text{f)} \int \frac{1}{2 + \sqrt{x+1}} dx
 \end{aligned}$$

$$\begin{aligned}
 \text{a)} \int \frac{1 + \sqrt{x+1}}{1 + \sqrt[3]{x+1}} dx &= \\
 & \boxed{t = \sqrt[6]{x+1} \rightarrow dt = \frac{1}{6\sqrt[6]{x+1}} dx \rightarrow dx = 6t^5 dt} \\
 &= \int \frac{1 + (\sqrt[6]{x+1})^3}{1 + (\sqrt[6]{x+1})^2} dx = \\
 &= \int \frac{1 + t^3}{1 + t^2} 6t^5 dt = 6 \int \frac{t^8 + t^5}{1 + t^2} dt = \\
 &= 6 \int \left(t^6 - t^4 + t^3 + t^2 - t + 1 + \frac{t}{t^2 + 1} + \frac{1}{t^2 + 1} \right) dt = \\
 &= 6 \left(\frac{t^7}{7} - \frac{t^5}{5} + \frac{t^4}{4} + \frac{t^3}{3} - \frac{t^2}{2} - t + \right. \\
 &\quad \left. + \frac{1}{2} \ln |t^2 + 1| + \operatorname{arctg} t \right) + k = \\
 &= 6 \left(\frac{\sqrt[6]{(x+1)^7}}{7} - \frac{\sqrt[6]{(x+1)^5}}{5} + \right. \\
 &\quad \left. + \frac{\sqrt[3]{(x+1)^4}}{4} + \frac{\sqrt{x+1}}{3} - \right. \\
 &\quad \left. - \frac{\sqrt[3]{(x+1)^2}}{2} - \sqrt[6]{x+1} + \right. \\
 &\quad \left. + \frac{1}{2} \ln |\sqrt[3]{(x+1)^2} + 1| + \right.
 \end{aligned}$$

$$+ \arctg \sqrt[6]{x+1} \Big) + k$$

$$t = \sqrt[6]{x} \rightarrow dt = \frac{1}{6\sqrt[5]{x^5}} dx \rightarrow dx = 6t^5 dt$$

$$\begin{aligned} \text{b) } \int \frac{\sqrt{x}}{1 + \sqrt[3]{x}} dx &= \int \frac{t^3}{1 + t^2} \cdot 6t^5 dt = \\ &= \int \frac{6t^8}{1 + t^2} dt = \\ &= 6 \int \left(t^6 - t^4 + t^2 - 1 + \frac{1}{t^2 + 1} \right) dt = \\ &= 6 \left(\frac{t^7}{7} - \frac{t^5}{5} + \frac{t^3}{3} - t + \arctg t \right) + k = \\ &= 6 \left(\frac{\sqrt[6]{x^7}}{7} - \frac{\sqrt[6]{x^5}}{5} + \frac{\sqrt{x}}{3} - \sqrt[6]{x} + \right. \\ &\quad \left. + \arctg \sqrt[6]{x} \right) + k \end{aligned}$$

$$t = \sqrt[4]{x+1} \rightarrow dt = \frac{1}{6\sqrt[3]{(x+1)^5}} dx \rightarrow dx = 6t^5 dt$$

$$\begin{aligned} \text{c) } \int \frac{1+x+\sqrt{x}}{(x+1)\sqrt[3]{x+1}} dx &= \\ &= \int \frac{t^6 + t^3}{t^6 \cdot t^2} \cdot 6t^5 dt = 6 \int (t^3 + 1) dt = \\ &= 6 \left(\frac{t^4}{4} + t \right) + k = \\ &= 6 \left(\frac{\sqrt[4]{(1+x)^4}}{4} + \sqrt[4]{1+x} \right) + k = \\ &= 6 \left(\frac{\sqrt[3]{(1+x)^2}}{4} + \sqrt[4]{1+x} \right) + k \end{aligned}$$

$$t = \sqrt[4]{x} \rightarrow dt = \frac{1}{4\sqrt[3]{x^3}} dx \rightarrow dx = 4t^3 dt$$

$$\begin{aligned} \text{d) } \int \frac{x + \sqrt{x}}{\sqrt{x} + \sqrt[4]{x}} dx &= \int \frac{t^4 + t^2}{t^2 + t} \cdot 4t^3 dt = \\ &= 4 \int \left(t^5 - t^4 + 2t^3 - 2t^2 + \right. \\ &\quad \left. + 2t - 2 + \frac{2}{t+1} \right) dt = 4 \left(\frac{t^6}{6} - \frac{t^5}{5} + \right. \\ &\quad \left. + \frac{2t^4}{4} - \frac{2t^3}{3} + \frac{2t^2}{2} - 2t + \right. \end{aligned}$$

$$\begin{aligned} &+ 2 \ln |t+1| \Big) + k = 4 \left(\frac{\sqrt[4]{x^6}}{6} - \right. \\ &- \frac{\sqrt[4]{x^5}}{5} + \frac{x}{2} - \frac{2\sqrt[4]{x^3}}{3} + \sqrt{x} - \\ &- 2\sqrt[4]{x} + 2 \ln |\sqrt[4]{x} + 1| \Big) + k \end{aligned}$$

$$t = \sqrt[4]{x} \rightarrow dt = \frac{1}{4\sqrt[3]{x^3}} dx \rightarrow dx = 4t^3 dt$$

$$\begin{aligned} \text{e) } \int \frac{1}{\sqrt{x}(1 + \sqrt[4]{x})} dx &= \\ &= \int \frac{1}{t^2(1+t)} \cdot 4t^3 dt = 4 \int \frac{t}{1+t} dt = \\ &= 4 \int \left(1 + \frac{-1}{1+t} \right) dt = \\ &= 4t - 4 \ln |t+1| + k = \\ &= 4\sqrt[4]{x} - 4 \ln |\sqrt[4]{x} + 1| + k \end{aligned}$$

$$t = \sqrt{x+1} \rightarrow dt = \frac{1}{2\sqrt{x+1}} dx \rightarrow dx = 2t dt$$

$$\begin{aligned} \text{f) } \int \frac{1}{2 + \sqrt{x+1}} dx &= \int \frac{1}{2+t} \cdot 2t dt = \\ &= 2 \int \left(1 + \frac{-2}{t+2} \right) dt = \\ &= 2t - 4 \ln |t+2| + k = \\ &= 2\sqrt{x+1} - 4 \ln |\sqrt{x+1} + 2| + k \end{aligned}$$

98 Calcula les integrals següents.

$$\text{a) } \int \frac{1}{(1-e^x)^2} dx \quad \text{c) } \int \frac{e^{3x}}{e^{2x} - 3e^x + 2} dx$$

$$\text{b) } \int \frac{1}{e^x + e^{-x} + 1} dx \quad \text{d) } \int \frac{1 + \sqrt{e^x}}{(1 - \sqrt[4]{e^x})^2} dx$$

$$t = e^x \rightarrow dt = e^x dx \rightarrow dx = \frac{dt}{t}$$

$$\begin{aligned} \text{a) } \int \frac{1}{(1-e^x)^2} dx &= \int \frac{1}{(1-t)^2} \cdot \frac{1}{t} dt = \\ &= \int \left(\frac{1}{t} - \frac{1}{t-1} + \frac{1}{(t-1)^2} \right) dt = \\ &= \ln |t| - \ln |t-1| - \frac{1}{t-1} + k = \\ &= x - \ln |e^x - 1| - \frac{1}{e^x - 1} + k \end{aligned}$$

$$\begin{aligned}
 \text{b) } \int \frac{1}{e^x + e^{-x} + 1} dx &= \\
 &= \int \frac{e^x}{e^{2x} + e^x + 1} dx \quad \boxed{t = e^x \rightarrow dt = e^x dx} \\
 &= \int \frac{1}{t^2 + t + 1} dt = \\
 &= \int \frac{1}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} dt = \\
 &= \frac{4}{3} \int \frac{1}{\left(\frac{2}{\sqrt{3}}t + \frac{1}{\sqrt{3}}\right)^2 + 1} dt = \\
 &= \frac{4}{3} \cdot \frac{\sqrt{3}}{2} \operatorname{arctg}\left(\frac{2}{\sqrt{3}}t + \frac{1}{\sqrt{3}}\right) + k = \\
 &= \frac{2}{\sqrt{3}} \operatorname{arctg}\left(\frac{2}{\sqrt{3}}e^x + \frac{1}{\sqrt{3}}\right) + k
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \int \frac{e^{3x}}{e^{2x} - 3e^x + 2} dx &= \quad \boxed{t = e^x \rightarrow dt = e^x dx} \\
 &= \int \frac{t^2}{t^2 - 3t + 2} dt = \\
 &= \int \left(-\frac{1}{t-1} + \frac{4}{t-2} + 1\right) dt = \\
 &= -\ln|t-1| + 4\ln|t-2| + t + k = \\
 &= -\ln|e^x - 1| + 4\ln|e^x - 2| + e^x + k
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \int \frac{1 + \sqrt[4]{e^x}}{(1 - \sqrt[4]{e^x})^2} dx &= \quad \boxed{t = \sqrt[4]{e^x} \rightarrow dt = \frac{\sqrt[4]{e^x}}{4} dx \rightarrow dx = \frac{4}{t} dt} \\
 &= 4 \int \frac{1 + t^2}{t(1-t)^2} dt = \\
 &= 4 \int \left(\frac{1}{t} + \frac{1}{1-t} + \frac{t+1}{(1-t)^2}\right) dt = \\
 &= 4 \left(\ln|t| - \ln|1-t| + \right. \\
 &\quad \left. + \int \frac{t-1+2}{(1-t)^2} dt \right) + k =
 \end{aligned}$$

$$\begin{aligned}
 &= 4 \left(\ln|t| - \ln|1-t| + \int \frac{-1}{1-t} dt + \right. \\
 &\quad \left. + \int \frac{2}{(1-t)^2} \right) + k = 4 \left(\ln|t| - \right. \\
 &\quad \left. - \ln|1-t| + \frac{2}{1-t} \right) + k = \\
 &= 4 \left(\ln|\sqrt[4]{e^x}| + \frac{2}{1 - \sqrt[4]{e^x}} \right) + k
 \end{aligned}$$

99 Integra mitjançant un canvi de variable.

...

$$\text{a) } \int 2 \sin x e^{-\cos x} dx$$

$$\text{b) } \int (1-x^2)^{-\frac{3}{2}} dx$$

$$\text{c) } \int x^5 \sqrt{1-x^2} dx$$

$$\text{d) } \int (1-(2x+1)^2)^{-\frac{1}{2}} dx$$

$$\begin{aligned}
 \text{a) } \int 2 \sin x e^{-\cos x} dx &= \quad \boxed{t = -\cos x \rightarrow dt = \sin x dx} \\
 &= \int 2e^t dt = \\
 &= 2e^t + k = 2e^{-\cos x} + k
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \int (1-x^2)^{-\frac{3}{2}} dx &= \quad \boxed{x = \sin t \rightarrow dx = \cos t dt} \\
 &= \int \frac{\cos t}{(1-\sin^2 t)^{\frac{3}{2}}} dt = \\
 &= \int \frac{\cos t}{\cos^3 t} dt = \int \frac{1}{\cos^2 t} dt = \operatorname{tg} t + k = \\
 &= \operatorname{tg}(\arcsin x) + k = \frac{x}{\sqrt{1-x^2}} + k
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \int x^5 \sqrt{1-x^2} dx &= \quad \boxed{t = \sin x \rightarrow dt = \cos x dx} \\
 &= \int \sin^5 t \cos^2 t dt = \\
 &= \int \sin t (1-\cos^2 t)^2 \cos^2 t dt = \quad \boxed{u = \cos t \rightarrow du = -\sin t dt} \\
 &= - \int (1-u^2)^2 u^2 du = \\
 &= - \int (u^2 - 2u^4 + u^6) du =
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{u^3}{3} + \frac{2u^5}{5} - \frac{u^7}{7} + k = \\
 &= -\frac{\cos^3 t}{3} + \frac{2\cos^5 t}{5} - \frac{\cos^7 t}{7} + k = \\
 &= \cos t \left(-\frac{1}{3}(1 - \sin^2 t) + \frac{2}{5}(1 - \sin^2 t)^2 - \frac{1}{7}(1 - \sin^2 t)^3 \right) + k = \\
 &= \sqrt{1 - x^2} \cdot \left(-\frac{1 - x^2}{3} + \frac{2(1 - x^2)^2}{5} - \frac{(1 - x^2)^3}{7} \right) + k
 \end{aligned}$$

$$t = 2x + 1 \rightarrow dt = 2dx$$

$$\begin{aligned}
 \text{d) } \int (1 - (2x + 1)^2)^{-\frac{1}{2}} dx &= \\
 &= \frac{1}{2} \int (1 - t^2)^{-\frac{1}{2}} dt = \frac{1}{2} \int \frac{1}{\sqrt{1 - t^2}} dt = \\
 &= \frac{1}{2} \arcsin t + k = \\
 &= \frac{1}{2} \arcsin(2x + 1) + k
 \end{aligned}$$

100 REpte. Tria un canvi de variable adequat per resoldre la integral

$$\int \frac{1 - \sqrt{x}}{\sqrt[3]{x}} dx \text{ i escriu-ne la resolució.}$$

Generalitza el resultat per resoldre les integrals de tipus $\int \frac{1 - \sqrt[n]{x}}{\sqrt[m]{x}} dx$.

Calculem m.c.m.(2, 3) = 6.

$$t = \sqrt[6]{x} \rightarrow dt = \frac{1}{6\sqrt[5]{x^5}} dx \rightarrow dx = 6t^5 dt$$

$$\begin{aligned}
 \int \frac{1 - \sqrt{x}}{\sqrt[3]{x}} dx &= \int \frac{1 - t^3}{t^2} dt = \\
 &= \int t^{-2} dt - \int t dt = -\frac{1}{t} - \frac{t^2}{2} + k = \\
 &= -\frac{1}{\sqrt[6]{x}} - \frac{\sqrt[3]{x}}{2} + k
 \end{aligned}$$

Generalitzant, si m.c.m.(m, n) = k, fem el canvi de variable $t = \sqrt[k]{x}$.

$$\int \frac{1 - \sqrt[n]{x}}{\sqrt[m]{x}} dx = \int \frac{1 - t^m}{t^n} dt = \int t^{-n} dt -$$

$$\begin{aligned}
 - \int t^{m-n} dt &= \frac{t^{-n+1}}{-n+1} - \frac{t^{m-n+1}}{m-n+1} + k = \\
 &= \frac{(\sqrt[k]{x})^{-n+1}}{-n+1} - \frac{(\sqrt[k]{x})^{m-n+1}}{m-n+1} + k
 \end{aligned}$$

Integrals de funcions

101 Calcula aquestes integrals.

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$$\text{a) } \int \frac{3x^2 - 5x + 7}{x^3 - 4x^2 + 4x} dx \quad \text{d) } \int \frac{5}{\sqrt{1 - 9x^2}} dx$$

$$\text{b) } \int \frac{x}{\sqrt{1 - 9x^4}} dx \quad \text{e) } \int \frac{x}{\sqrt{1 + x^2}} dx$$

$$\text{c) } \int \frac{3x^2 - 7x + 4}{2x - 3} dx \quad \text{f) } \int \sqrt{1 - x^2} dx$$

$$\begin{aligned}
 \text{a) } \int \frac{3x^2 - 5x + 7}{x^3 - 4x^2 + 4x} dx &= \\
 &= \int \left(\frac{9}{2(x - 2)^2} + \frac{5}{4(x - 2)} + \frac{7}{4x} \right) dx = \\
 &= \frac{-9}{2(x - 2)} + \frac{5}{4} \ln|x - 2| + \frac{7}{4} \ln|x| + k
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \int \frac{x}{\sqrt{1 - 9x^2}} dx &= \\
 &= -\frac{1}{18} \int \frac{-18x}{\sqrt{1 - 9x^2}} dx = \\
 &= -\frac{\sqrt{1 - 9x^2}}{9} + k
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \int \frac{3x^2 - 7x + 4}{2x - 3} dx &= \\
 &= \int \left(\frac{3}{2}x - \frac{5}{4} + \frac{1}{4(2x - 3)} \right) dx = \\
 &= \frac{3x^2}{4} - \frac{5x}{4} + \frac{1}{8} \ln|2x - 3| + k
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \int \frac{5}{\sqrt{1 - 9x^2}} dx &= \frac{5}{3} \int \frac{3}{\sqrt{1 - (3x)^2}} dx = \\
 &= \frac{5}{3} \arcsin(3x) + k
 \end{aligned}$$

$$\text{e) } \int \frac{x}{\sqrt{1 + x^2}} dx = \sqrt{1 + x^2} + k$$

$$x = \sin t \rightarrow dx = \cos t dt$$

$$\text{f) } \int \sqrt{1 - x^2} dx = \int \sqrt{1 - \sin^2 t} \cos t dt =$$

$$\begin{aligned} u = \cos t &\rightarrow du = -\sin t dt \\ dv = \cos t &\rightarrow v = \sin t \end{aligned}$$

$$\begin{aligned} &= \int \cos^2 t dt = \cos t \sin t + \int \sin^2 t dt = \\ &= \cos t \sin t + \int (1 - \cos^2 t) dt = \\ &= \cos t \sin t + t - \int \cos^2 t dt \\ \int \cos^2 t dt &= \\ &= \cos t \sin t + t - \int \cos^2 t dt \rightarrow \\ \rightarrow \int \cos^2 t dt &= \frac{1}{2}(\cos t \sin t + t) + k \\ \int \sqrt{1-x^2} dx &= \frac{x\sqrt{1-x^2} + \arcsin x}{4} + k \end{aligned}$$

102 Troba les integrals següents.

a) $\int \frac{x^3 + 4x^2 - 10x + 7}{x^3 - 7x - 6} dx$ c) $\int \frac{1}{x\sqrt{1-x}} dx$

b) $\int \frac{1}{x^2 - 7x + 10} dx$ d) $\int \frac{x+3}{4x^2+8} dx$

a) $\int \frac{x^3 + 4x^2 - 10x + 7}{x^3 - 7x - 6} dx =$
 $= \int \left(1 + \frac{2}{x-3} - \frac{5}{x+1} + \frac{7}{x+2} \right) dx =$
 $= x + 2 \ln|x-3| - 5 \ln|x+1| +$
 $+ 7 \ln|x+2| + k$

b) $\int \frac{1}{x^2 - 7x + 10} dx =$
 $= \int \left(\frac{1}{3(x-5)} - \frac{1}{3(x-2)} \right) dx =$
 $= \frac{1}{3} \ln|x-5| - \frac{1}{3} \ln|x-2| + k$

$$t = \sqrt{1-x} \rightarrow dt = \frac{-1}{2\sqrt{1-x}} dx$$

c) $\int \frac{1}{x\sqrt{1-x}} dx = \int \frac{-2}{1-t^2} dt =$
 $= \int \frac{-1}{1-t} dt - \int \frac{1}{1+t} dt =$
 $= \ln|1-t| - \ln|1+t| + k =$
 $= \ln|1-\sqrt{1-x}| - \ln|1+\sqrt{1-x}| +$

d) $\int \frac{x+3}{4x^2+8} dx =$
 $= \frac{1}{8} \int \frac{2x}{x^2+2} dx + \frac{1}{4} \int \frac{3}{x^2+2} dx =$
 $= \frac{1}{8} \ln|x^2+2| + \frac{3}{4\sqrt{2}} \operatorname{arctg} \frac{x}{\sqrt{2}} + k$

103 Calcula aquestes integrals.

a) $\int \operatorname{cosec} x dx$ d) $\int \frac{1}{x\sqrt{1-(\ln x)^2}} dx$

b) $\int \sec x dx$ e) $\int \frac{1+e^{3x}}{e^{2x}} dx$

c) $\int e^{x^2-5x} (2x-5) dx$ f) $\int \sqrt{e^x-1} dx$

$$t = \cos x \rightarrow dt = -\sin x dx$$

a) $\int \operatorname{cosec} x dx = \int \frac{1}{\sin x} dx =$
 $= \int \frac{-1}{1-t^2} dt = \frac{-1}{2} \ln|1+t| +$
 $+ \frac{1}{2} \ln|1-t| + k = -\frac{1}{2} \ln|1| +$
 $+ \cos x| + \frac{1}{2} \ln|1-\cos x| + k$

$$t = \sin x \rightarrow dt = \cos x dx$$

b) $\int \sec x dx = \int \frac{1}{\cos x} dx =$
 $= \int \frac{1}{1-t^2} dt =$
 $= \frac{1}{2} \ln|1+t| - \frac{1}{2} \ln|1-t| + k =$
 $= \frac{1}{2} \ln|1+\sin x| - \frac{1}{2} \ln|1-\sin x| + k$

c) $\int e^{x^2-5x} (2x-5) dx = e^{x^2-5x} + k$

d) $\int \frac{1}{x\sqrt{1-(\ln x)^2}} dx = \arcsin(\ln x) + k$

e) $\int \frac{1+e^{3x}}{e^{2x}} dx = \int \frac{1}{e^{2x}} dx + \int e^x dx =$
 $= \frac{-e^{-2x}}{2} + e^x + k$

$$t = e^x - 1 \rightarrow dt = e^x dx$$

$$\begin{aligned} \text{f) } \int \sqrt{e^x - 1} dx &= \int (t + 1) \sqrt{t} dt = \\ &= \frac{2}{5} t^{\frac{5}{2}} + \frac{2}{3} t^{\frac{3}{2}} + k = \\ &= \sqrt{e^x - 1} \left(\frac{2}{5} (e^x - 1)^2 + \frac{2}{3} (e^x - 1) \right) + k \end{aligned}$$

104 ●●● Calcula el resultat de les integrals.

$$\begin{array}{ll} \text{a) } \int (x - 2) e^{3x} dx & \text{d) } \int \frac{5e^{2x} - e^x}{e^{2x} - 1} dx \\ \text{b) } \int \frac{7}{x^3 \sqrt{\ln x}} dx & \text{e) } \int \sin^2 x dx \\ \text{c) } \int \frac{(\ln x)^2 + x}{x} dx & \text{f) } \int \cos^2 x dx \end{array}$$

$$\begin{array}{l} u = x - 2 \rightarrow du = dx \\ dv = e^{3x} dx \rightarrow v = \frac{e^{3x}}{3} \end{array}$$

$$\begin{aligned} \text{a) } \int (x - 2) e^{3x} dx &= \\ &= \frac{1}{3} (x - 2) e^{3x} - \int \frac{e^{3x}}{3} dx = \\ &= \frac{1}{3} (x - 2) e^{3x} - \frac{e^{3x}}{9} + k \\ \text{b) } \int \frac{7}{x^3 \sqrt{\ln x}} dx &= \frac{21}{2} \sqrt[3]{(\ln x)^2} + k \\ \text{c) } \int \frac{(\ln x)^2 + x}{x} dx &= \\ &= \int \frac{(\ln x)^2}{x} dx + \int dx = \\ &= \frac{(\ln x)^3}{3} + x + k \end{aligned}$$

$$t = e^x \rightarrow dt = e^x dx$$

$$\begin{aligned} \text{d) } \int \frac{5e^{2x} - e^x}{e^{2x} - 1} dx &= \int \frac{5t - 1}{t^2 - 1} dt = \\ &= \int \frac{2}{t - 1} dt + \int \frac{3}{t + 1} dt = \\ &= 2 \ln |t - 1| + 3 \ln |t + 1| + k = \\ &= 2 \ln |e^x - 1| + 3 \ln |e^x + 1| + k \\ \text{e) } \int \sin^2 x dx &= \int \frac{1 - \cos(2x)}{2} dx = \end{aligned}$$

$$= \frac{x}{2} - \frac{\sin(2x)}{4} + k$$

$$\begin{aligned} \text{f) } \int \cos^2 x dx &= \int \frac{1 + \cos(2x)}{2} dx = \\ &= \frac{x}{2} + \frac{\sin(2x)}{4} + k \end{aligned}$$

105 ●●● Determina aquestes integrals.

$$\begin{array}{ll} \text{a) } \int 2^x \cos x dx & \text{d) } \int \sin^2 x \cos x dx \\ \text{b) } \int \frac{\ln x + 3}{x(\ln x - 1)} dx & \text{e) } \int \sin^3 x \cos x dx \\ \text{c) } \int \cos(\ln x) dx & \text{f) } \int \sin^3 x dx \end{array}$$

$$\begin{array}{l} u = 2^x \rightarrow du = 2^x \ln 2 dx \\ dv = \cos x dx \rightarrow v = \sin x \end{array}$$

$$\begin{aligned} \text{a) } \int 2^x \cos x dx &= \\ &= 2^x \sin x - \ln 2 \int 2^x \sin x dx = \\ &= 2^x \sin x + \ln 2 \cdot 2^x \cos x - \\ &\quad - (\ln 2)^2 \int 2^x \cos x dx \\ \int 2^x \cos x dx &= \frac{2^x \sin x + \ln 2 \cdot 2^x \cos x}{1 + (\ln 2)^2} \end{aligned}$$

$$\begin{array}{l} u = 2^x \rightarrow du = 2^x \ln 2 dx \\ dv = \sin x dx \rightarrow v = -\cos x \end{array}$$

$$\begin{aligned} &= 2^x \sin x - \ln 2 \int 2^x \sin x dx = \\ &= 2^x \sin x + \ln 2 \cdot 2^x \cos x - \\ &\quad - (\ln 2)^2 \int 2^x \cos x dx \\ \int 2^x \cos x dx &= \frac{2^x \sin x + \ln 2 \cdot 2^x \cos x}{1 + (\ln 2)^2} \end{aligned}$$

$$t = \ln x \rightarrow dt = \frac{dx}{x}$$

$$\begin{aligned} \text{b) } \int \frac{\ln x + 3}{x(\ln x - 1)} dx &= \int \frac{t + 3}{t - 1} dt = \\ &= t + 4 \ln |t - 1| + k = \\ &= \ln |x| + 4 \ln |\ln x - 1| + k \end{aligned}$$

$$\begin{array}{l} t = \ln x \rightarrow dt = \frac{dx}{x} \\ u = e^t \rightarrow du = e^t dt \\ dv = \cos t dt \rightarrow v = \sin t \end{array}$$

$$\begin{aligned} \text{c) } \int \cos(\ln x) dx &= \int e^t \cos t dt = \\ &= e^t \sin t - \int e^t \sin t dt = \end{aligned}$$

$$\begin{array}{l} u = e^t \rightarrow du = e^t dt \\ dv = \sin t dt \rightarrow v = -\cos t \end{array}$$

$$= e^t \sin t + e^t \cos t - \int e^t \cos t dt$$

$$\int \cos(\ln x) dx = \frac{e^t \sin t + e^t \cos t}{2} + k =$$

$$= \frac{x \sin(\ln x) + x \cos(\ln x)}{2} + k$$

d) $\int \sin^2 x \cos x dx = \frac{\sin^3 x}{3} + k$

e) $\int \sin^3 x \cos x dx = \frac{\sin^4 x}{4} + k$

f) $\int \sin^3 x dx = \int (1 - \cos^2 x) \sin x dx =$

$$= \int \sin x dx - \int \cos^2 x \sin x dx =$$

$$= -\cos x + \frac{\cos^3 x}{3} + k$$

106 INVESTIGA. Resol aquesta integral.

$$\int \frac{\sin x - \sin^3 x}{\cos^4 x - \cos^2 x \sin^2 x} dx$$

$$\int \frac{\sin x - \sin^3 x}{\cos^4 x + \cos^2 x \sin^2 x} dx =$$

$$= \int \frac{\sin x (1 - \sin^2 x)}{\cos^2 x (\cos^2 x + \sin^2 x)} dx =$$

$$= \int \sin x dx = -\cos x + k$$

107 Calcula les integrals.

a) $\int x^3 \sqrt{2x+1} dx$

b) $\int \frac{-2x+6}{x^3-2x^2-x+2} dx$

c) $\int \frac{\sin(2x)}{\sqrt{1+\cos(2x)}} dx$

d) $\int \frac{1}{x^4 \sqrt{x^2-1}} dx$

a) $\int x^3 \sqrt{2x+1} dx =$

$$= \frac{1}{2} \int \left(\frac{t-1}{2} \right)^3 \sqrt{t} dt =$$

$$= \frac{1}{16} \int (t-1)^3 \sqrt{t} dt =$$

$$= \frac{1}{16} \int (t^{7/2} - 3t^{5/2} + 3t^{3/2} - t^{1/2}) dt =$$

$$= \frac{1}{16} \left(\frac{2}{9} t^{9/2} - \frac{6}{7} t^{7/2} + \frac{6}{5} t^{5/2} - \frac{2}{3} t^{3/2} \right) +$$

$$+ k = \frac{1}{16} \left(\frac{2}{9} (2x+1)^{9/2} - \right.$$

$$- \frac{6}{7} (2x+1)^{7/2} + \frac{6}{5} (2x+1)^{5/2} -$$

$$\left. - \frac{2}{3} (2x+1)^{3/2} \right) + k$$

b) $\int \frac{-2x+6}{x^3-2x^2-x+2} dx =$

$$= \int \left(-\frac{2}{x-1} + \frac{4}{3(x+1)} + \frac{2}{3(x-2)} \right) dx =$$

$$= -2 \ln|x-1| + \frac{4}{3} \ln|x+1| +$$

$$+ \frac{2}{3} \ln|x-2| + k$$

c) $\int \frac{\sin(2x)}{\sqrt{1+\cos(2x)}} dx = -\frac{1}{2} \int \frac{1}{\sqrt{t}} dt =$

$$= -\sqrt{t} + k = -\sqrt{1+\cos(2x)} + k$$

d) $\int \frac{1}{x^4 \sqrt{x^2-1}} dx =$

$$= \int \frac{\operatorname{tg} t \operatorname{sect}}{\sec^4 t \cdot \sqrt{\sec^2 t - 1}} dt =$$

$$= \int \frac{\operatorname{tg} t \operatorname{sect}}{\sec^4 t \cdot \operatorname{tg} t} dt = \int \frac{1}{\sec^3 t} dt =$$

$$= \int \cos^3 t dt = \int (1 - \sin^2 x) \cos x dx =$$

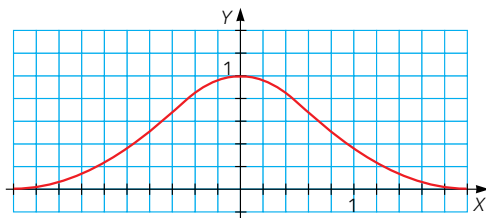
$$= \int \cos x dx - \int \sin^2 x \cos x dx =$$

$$= \sin x - \frac{\sin^3 x}{3} + k$$

108 REpte. Determina la funció $f(x)$ que compleix l'equació $f'(x) + 2x f(x) = 0$, sabent que $f(0) = 1$. Representa gràficament la funció

i calcula la tangent en el punt d'abscissa 0 de la corba.

$$f(x) = e^{-x^2}; f'(x) = -2xe^{-x^2}$$



$$y - f(0) = f'(0)(x - 0) \rightarrow y - 1 = 0 \rightarrow y = 1$$

109 Calcula aquestes integrals.

...
a) $\int 4 \sin(3x) \sin(2x) dx$

b) $\int -3 \sin(2x) \cos x dx$

a) $\int 4 \sin(3x) \sin(2x) dx =$
 $= 4 \int \frac{1}{2} (\cos(-x) - \cos(5x)) dx =$
 $= -2 \sin(-x) - \frac{2}{5} \sin(5x) + k$

b) $\int -3 \sin(2x) \cos x dx =$
 $= -3 \int \frac{1}{2} (\sin x + \sin(3x)) dx =$
 $= \frac{3}{2} \cos x + \frac{1}{2} \cos(3x) + k$

110 Troba, amb el mètode que consideris més adequat, aquestes integrals.

...
a) $\int \frac{1}{\sqrt{x+2} + \sqrt{x-2}} dx$

b) $\int \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1-x^4}} dx$

c) $\int \frac{x^4 + 5\sqrt[3]{x} - 3x\sqrt{x}}{4x} dx$

d) $\int (x - x^{-3}) \sqrt{x\sqrt{x}\sqrt{x}} dx$

a) $\int \frac{1}{\sqrt{x+2} + \sqrt{x-2}} dx =$

$$= \int \frac{1}{4} (\sqrt{x+2} - \sqrt{x-2}) dx =$$

$$= \frac{1}{6} \sqrt{(x+2)^3} - \frac{1}{6} \sqrt{(x-2)^3} + k$$

b) $\int \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1-x^4}} dx =$

$$= \int \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{(1+x^2)(1-x^2)}} dx =$$

$$= \int \frac{1}{\sqrt{1+x^2}} dx + \int \frac{1}{\sqrt{1-x^2}} dx =$$

$$= \operatorname{arcsinh} x + \operatorname{arcsin} x + k$$

c) $\int \frac{x^4 + 5\sqrt[3]{x} - 3x\sqrt{x} - 2}{4x} dx =$

$$= \frac{1}{4} \int (x^3 + 5x^{-\frac{2}{3}} - 3\sqrt{x} - 2x^{-1}) dx =$$

$$= \frac{1}{4} \left(\frac{1}{4} x^4 + 15\sqrt[3]{x} - 2\sqrt{x^3} - 2 \ln |x| \right) + k$$

d) $\int (x - x^{-3}) \sqrt{x\sqrt{x}\sqrt{x}} dx =$

$$= \int (x - x^{-3}) \cdot x^{\frac{1}{2}} \cdot x^{\frac{1}{4}} \cdot x^{\frac{1}{8}} dx =$$

$$= \int (x - x^{-3}) \cdot x^{\frac{7}{8}} dx =$$

$$= \int \left(x^{\frac{15}{8}} - x^{-\frac{17}{8}} \right) dx =$$

$$= \frac{8}{23} x^{\frac{23}{8}} + \frac{8}{9} x^{-\frac{9}{8}} + k =$$

$$= \frac{8}{23} \sqrt[8]{x^{23}} + \frac{8}{9 \sqrt[8]{x^9}} + k$$

111 Integra aquestes funcions.

...
a) $f(x) = \frac{3}{\sqrt[3]{x^2}}$

b) $g(x) = \frac{1}{4+x^2}$

c) $h(x) = \frac{\cos x}{1+\sin^2 x}$

a) $\int f(x) dx = \int \frac{3}{\sqrt[3]{x^2}} dx = 3 \int x^{-\frac{2}{3}} dx =$
 $= 9\sqrt[3]{x} + k$

b) $\int g(x) dx = \int \frac{1}{4+x^2} dx =$

$$= \frac{1}{4} \int \frac{1}{1 + \left(\frac{x}{2}\right)^2} = \frac{1}{2} \operatorname{arctg}\left(\frac{x}{2}\right) + k$$

$$t = \sin x \rightarrow dt = \cos x dx$$

$$\begin{aligned} \text{c) } \int h(x) dx &= \int \frac{\cos x}{1 + \sin^2 x} dx = \\ &= \int \frac{1}{1 + t^2} dt = \operatorname{arctg} t + k = \\ &= \operatorname{arctg}(\sin x) + k \end{aligned}$$

112 Calcula les integrals següents.

$$\text{a) } \int \frac{\sin(5x)}{\cos^2(5x)} dx \quad \text{c) } \int x^2 \sqrt{3x^3 + 7} dx$$

$$\text{b) } \int \frac{3x + 2}{x^2 + 8x + 7} dx \quad \text{d) } \int \left(\frac{\sqrt{x}}{3x} - \frac{5x}{\sqrt[3]{x}} \right) dx$$

$$t = \cos(5x) \rightarrow dt = -5 \sin(5x) dx$$

$$\begin{aligned} \text{a) } \int \frac{\sin(5x)}{\cos^2(5x)} dx &= -\frac{1}{5} \int \frac{1}{t^2} dt = \\ &= \frac{1}{5t} + k = \frac{1}{5 \cos(5x)} + k \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{3x + 2}{x^2 + 8x + 7} dx &= \\ &= \int \left(\frac{19}{6(x + 7)} - \frac{1}{6(x + 1)} \right) dx = \\ &= \frac{19}{6} \ln|x + 7| - \frac{1}{6} \ln|x + 1| + k \end{aligned}$$

$$t = 3x^3 + 7 \rightarrow dt = 9x^2 dx$$

$$\begin{aligned} \text{c) } \int x^2 \cdot \sqrt{3x^3 + 7} dx &= \frac{1}{9} \int \sqrt{t} dt = \\ &= \frac{1}{9} \cdot \frac{2}{3} \cdot t^{\frac{3}{2}} + k = \frac{2}{27} \sqrt{(3x^3 + 7)^3} + k \end{aligned}$$

$$\begin{aligned} \text{d) } \int \left(\frac{\sqrt{x}}{3x} - \frac{5x}{\sqrt[3]{x}} \right) dx &= \\ &= \int \left(\frac{1}{3} x^{-\frac{1}{2}} - 5x^{\frac{2}{3}} \right) dx = \\ &= \frac{2}{3} x^{\frac{1}{2}} - 3x^{\frac{5}{3}} + k = \frac{2\sqrt{x}}{3} - 5\sqrt[3]{x^5} + k \end{aligned}$$

113 Troba aquestes integrals.

$$\text{a) } \int \frac{1 + (\ln x)^3}{x(\ln^4 x + \ln^2 x)} dx$$

$$\text{b) } \int e^x (e^x \sin e^x) dx$$

$$t = \ln x \rightarrow dt = \frac{1}{x} dx$$

$$\begin{aligned} \text{a) } \int \frac{1 + (\ln x)^3}{x(\ln^4 x + \ln^2 x)} dx &= \int \frac{1 + t^3}{t^4 + t^2} dt = \\ &= \int \left(\frac{t-1}{t^2+1} + \frac{1}{t^2} \right) dt = \\ &= \int \left(\frac{t}{t^2+1} - \frac{1}{t^2+1} + \frac{1}{t^2} \right) dt = \\ &= \frac{1}{2} \ln|t^2 + 1| - \operatorname{arctg} t - \frac{1}{t} + k = \\ &= \frac{1}{2} \ln|(\ln x)^2 + 1| - \operatorname{arctg}(\ln x) - \frac{1}{\ln x} + k \end{aligned}$$

$$t = e^x \rightarrow dt = e^x dx$$

$$\begin{aligned} \text{b) } \int e^x (e^x \cdot \sin(e^x)) dx &= \int t \sin t dt = \\ &= -t \cos t - \int \cos t dt = -t \cos t - \sin t + \\ &+ k = -e^x \cos e^x - \sin e^x + k \end{aligned}$$

114 REpte. Resol en funció de a.

$$\begin{aligned} &\int \frac{a}{\sqrt{a^2 - x^2}} dx \\ \text{Si } a = 0: &\int \frac{a}{\sqrt{a^2 - x^2}} dx = k \\ \text{Si } a \neq 0: &\int \frac{a}{\sqrt{a^2 - x^2}} dx = \\ &= \int \frac{1}{\sqrt{1 - \left(\frac{x}{a}\right)^2}} dx = a \cdot \arcsin\left(\frac{x}{a}\right) + k \end{aligned}$$

115 Determina, en funció del paràmetre a, la integral següent.

$$\begin{aligned} &\int \frac{3}{x^2 - (a+2)x + 2a} dx \\ \text{Si } a \neq 2: &\int \frac{3dx}{x^2 - (a+2)x + 2a} = \\ &= \int \frac{3dx}{(x-a)(x-2)} = \\ &= -\frac{3}{2-a} \left(\int \frac{dx}{x-a} - \int \frac{dx}{x-2} \right) = \\ &= -\frac{3}{2-a} (\ln|x-a| - \ln|x-2|) + k \end{aligned}$$

$$\begin{aligned}\text{Si } a = 2: \int \frac{3}{(x-2)^2} dx &= \\ &= \int 3(x-2)^{-2} dx = \frac{-3}{x-2} + k\end{aligned}$$

116 ... Calcula la integral.

$$\begin{aligned}&\int (1 - \cos^2 x) \sin(2x) e^{\cos^2 x} dx \\&\int (1 - \cos^2 x) \cdot \sin(2x) \cdot e^{\cos^2 x} dx = \\&= \int (1 - \cos^2 x) \cdot 2 \sin x \cos x \cdot e^{\cos^2 x} dx = \\&\quad \boxed{t = \cos^2 x \rightarrow dt = -2 \cos x \sin x dx} \\&\quad \downarrow \\&= - \int (1 - t) e^t dt = - \int e^t dt + \int t e^t dt = \\&\quad \boxed{u = t \rightarrow du = dt} \\&\quad \boxed{dv = e^t dt \rightarrow v = e^t} \\&\quad \downarrow \\&= -e^t + t e^t - \int e^t dt = -e^t + t e^t - e^t + k = \\&= e^{\cos^2 x} (\cos^2 x - 2) + k\end{aligned}$$



117 ... **REpte.** Troba la família de funcions tals que el pendent de les rectes tangents a les gràfiques es troba substituint el valor de l'abscissa en la funció $f(x) = x e^{2x}$.

De manera particular, digues quina és la funció que passa pel punt $A(0, 2)$.

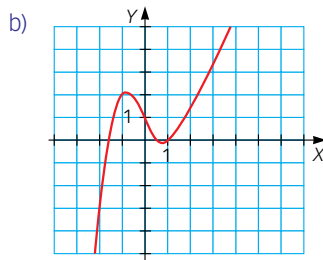
$$\begin{aligned}F(x) &= \int f(x) dx = \int x e^{2x} dx = \\&= \frac{1}{2} x e^{2x} - \frac{1}{2} \int e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + k \\F(0) &= 2 \rightarrow -\frac{1}{4} + k = 2 \rightarrow k = \frac{9}{4} \\&\text{La funció que compleix les condicions és} \\F(x) &= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + \frac{9}{4}.\end{aligned}$$

118 ... D'una funció f en coneixem la derivada.

$$f'(x) = \begin{cases} 3x^2 - 2 & \text{si } x \leq 1 \\ 1 + \ln x & \text{si } x > 1 \end{cases}$$

- a) Determina l'expressió de la funció f sabent que passa pel punt $P(-1, 2)$.
- b) Representa gràficament la funció f .

$$\begin{aligned}\text{a) } f(x) &= \int f'(x) dx = \\&= \begin{cases} \int (3x^2 - 2) dx & \text{si } x \leq 1 \\ \int (1 + \ln x) dx & \text{si } x > 1 \end{cases} \\f_1(x) &= \int (3x^2 - 2) dx = \\&= x^3 - 2x + k \rightarrow f_1(-1) = 2 \rightarrow \\&\rightarrow -1 + 2 + k = 2 \rightarrow k = 1 \rightarrow \\&\rightarrow f_1(x) = x^3 - 2x + 1 \\f_2(x) &= \int (1 + \ln x) dx = x + \int \ln x dx = \\&= x + x \ln x - \int dx = x + x \ln x - x = \\&= x \ln x + k \\F &\text{ ha de ser contínua en } x = 1, \text{ llavors:} \\f_1(1) &= f_2(1) \rightarrow 1 - 2 + 1 = 1 \cdot \ln 1 + k \rightarrow \\&\rightarrow k = 0 \rightarrow f_2(x) = x \ln x \\f(x) &= \int f'(x) dx = \\&= \begin{cases} x^3 - 2x + 1 & \text{si } x \leq 1 \\ x \ln x & \text{si } x > 1 \end{cases}\end{aligned}$$



119 ... D'una funció derivable f en sabem la derivada.

$$f'(x) = \begin{cases} \frac{-1}{\sqrt{-x}} & \text{si } x \leq -1 \\ 5x^4 - 6x^2 & \text{si } x > -1 \end{cases}$$

Troba l'expressió de f sabent que passa pel punt $P(2, 15)$.

$$\begin{aligned}f(x) &= \int f'(x) dx = \\&= \begin{cases} \int \frac{-1}{\sqrt{-x}} dx & \text{si } x \leq -1 \\ \int (5x^4 - 6x^2) dx & \text{si } x > -1 \end{cases}\end{aligned}$$

$$f_1(x) = \int \frac{-1}{\sqrt{-x}} dx = 2\sqrt{-x} + k$$

$$f_2(x) = \int (5x^4 - 6x^2) dx = x^5 - 2x^3 + k \rightarrow$$

$$\rightarrow f_2(2) = 15 \rightarrow 32 - 16 + k = 15 \rightarrow$$

$$\rightarrow k = -1 \rightarrow f_2(x) = x^5 - 2x^3 - 1$$

F ha de ser contínua en $x = -1$, llavors:

$$f_1(-1) = f_2(-1) \rightarrow 2 + k = 0 \rightarrow$$

$$\rightarrow k = -2 \rightarrow f_1(x) = 2\sqrt{-x} - 2$$

$$f(x) = \begin{cases} 2\sqrt{-x} - 2 & \text{si } x \leq -1 \\ x^5 - 2x^3 - 1 & \text{si } x > -1 \end{cases}$$

$$Q = \int I dt$$

En què Q és la càrrega elèctrica total i I és el corrent elèctric.

Quina és la càrrega total si el corrent és $I = 2t^2 - 5t$ A?

$Q(t) = \frac{2t^3}{3} - \frac{5t^2}{2} + C$, amb C una constant. Per obtenir la càrrega elèctrica en l'interval de temps $[t_1, t_2]$, avaluem la funció $Q(t)$ en t_2 i t_1 , i restem els resultats.

$$\begin{aligned} Q &= Q(t_2) - Q(t_1) = \\ &= \left(\frac{2t_2^3}{3} - \frac{5t_2^2}{2} + C \right) - \left(\frac{2t_1^3}{3} - \frac{5t_1^2}{2} + C \right) = \\ &= \frac{2t_2^3}{3} - \frac{5t_2^2}{2} - \frac{2t_1^3}{3} + \frac{5t_1^2}{2} \end{aligned}$$

120 INVESTIGA. Troba la primitiva de la

funció $f(x) = \frac{-3}{(3x+a)^2}$ en els casos

següents.

a) La recta $y = 4$ i l'eix d'ordenades són asymptotes de la funció.

b) La recta $x = 1$ i l'eix d'abscisses són asymptotes de la funció.

$$\begin{aligned} \int \frac{-3}{(3x+a)^2} dx &= - \int 3(3x+a)^{-2} dx = \\ &= \frac{1}{3x+a} + k \end{aligned}$$

a) Perquè $y = 4$ sigui asymptota, k ha de valer 4.

Perquè l'eix d'abscisses ($y = 0$) sigui asymptota, k ha de valer 0.

b) Perquè $x = 1$ sigui asymptota, a ha de valer -3 .

Perquè l'eix d'ordenades ($x = 0$) sigui asymptota, a ha de valer 0.

3. Fa servir el concepte d'integral per resoldre problemes de la vida quotidiana

121 MATEMÀTIQUES I... ELECTRICITAT.

El corrent elèctric es defineix com el flux de càrrega a través d'un punt en un circuit en un determinat interval de temps. La càrrega elèctrica en un sistema es pot calcular com la integral de corrent elèctric en funció del temps:

122 MATEMÀTIQUES I... FÍSICA. En física,

l'impuls es defineix com la quantitat de moviment d'un objecte.

Matemàticament, l'impuls que es produeix sobre una partícula de massa m en aplicar-li una força F d'intensitat variable es pot calcular mitjançant la integral:

$$\int F(t) dt$$

Calcula l'expressió de l'impuls que es produeix sobre una partícula quan se li aplica una força variable

$$F(t) = \frac{t^2}{3} + 2.$$

Impuls = $I(t) = \frac{t^3}{9} + 2t + C$, amb C una constant. Per obtenir l'impuls en l'interval de temps $[t_1, t_2]$, avaluem la funció $I(t)$ en t_2 i t_1 , i restem els resultats.

$$\begin{aligned} I &= I(t_2) - I(t_1) = \\ &= \left(\frac{t_2^3}{9} + 2t_2 + C \right) - \left(\frac{t_1^3}{9} + 2t_1 + C \right) = \\ &= \frac{t_2^3}{9} + 2t_2 - \frac{t_1^3}{9} - 2t_1 \end{aligned}$$

123 MATEMÀTIQUES I... FÍSICA. Com que
 ●●● l'acceleració és la variació de velocitat per

unitat de temps, podem calcular la velocitat a partir de l'acceleració

mitjançant una integral: $v = \int a dt$.

A la vegada, la velocitat és la variació de l'espai recorregut per unitat de temps;

per tant, $s = \int v dt$.

- a) Amb aquestes dades, dedueix les equacions per a la velocitat i l'espai recorregut en el moviment uniformement accelerat amb acceleració constant $a = k$.

- b) Quines serien les equacions si l'acceleració fos $a(t) = 1 + t$?

a) $v = \int k dt = kt + C_1$

$$s = \int (kt + C_1) dt = k \frac{t^2}{2} + C_1 t + C_2$$

b) $v = \int (1 + t) dt = t + \frac{t^2}{2} + C_1$

$$s = \int \left(t + \frac{t^2}{2} + C_1 \right) dt = \frac{t^2}{2} + \frac{t^3}{6} + C_1 t + C_2$$

124 MATEMÀTIQUES I... CANVI

●●● **CLIMÀTIC.** Se sap que l'augment de temperatura afecta negativament les poblacions d'algues en els esculls de coral.

El mesurament d'aquest impacte depèn de la temperatura. En una determinada població d'algues, la funció que en mesura el creixement ve donada per $c(t) = (t^2 - 1)\sqrt{e^t}$, en què t és la temperatura de l'aigua en aquell moment. Per calcular el nombre total d'algues se sumen tots els increments en cada moment del temps, és a dir, es calcula la integral. Quina serà la funció que determina el nombre d'algues segons la temperatura?



$$\int c(t) dt = 2e^{\frac{t}{2}} t^2 - 8e^{\frac{t}{2}} t + 14e^{\frac{t}{2}} + C$$

Per obtenir el nombre d'algues en l'interval de temperatura $[t_1, t_2]$, la funció obtinguda en t_2 i t_1 , i els resultats.

$$\begin{aligned} & \left(2e^{\frac{t_2}{2}} t_2^2 - 8e^{\frac{t_2}{2}} t_2 + 14e^{\frac{t_2}{2}} + C \right) - \\ & - \left(2e^{\frac{t_1}{2}} t_1^2 - 8e^{\frac{t_1}{2}} t_1 + 14e^{\frac{t_1}{2}} + C \right) = \\ & = 2e^{\frac{t_2}{2}} t_2^2 - 8e^{\frac{t_2}{2}} t_2 + 14e^{\frac{t_2}{2}} - 2e^{\frac{t_1}{2}} t_1^2 + \\ & + 8e^{\frac{t_1}{2}} t_1 - 14e^{\frac{t_1}{2}} \end{aligned}$$

PROBLEMES APARENTEMENT DIFERENTS

- 125** Calcula la integral d'aquesta funció.

$$f(x) = e^x \sqrt[3]{(e^x + 1)^2}$$

$$\int f(x) dx = \frac{3}{5} (e^x + 1) \sqrt[3]{(e^x + 1)^2} + C$$

- 126** La potència generada en cada instant en desplaçar una caixa segueix la funció $P(t) = e^t \sqrt[3]{(e^t + 1)^2}$.

Calcula el treball, $W(t)$, en cada instant sabent que la potència és la derivada d'aquest treball respecte del temps.

$$W(t) = \int P(t) dt = \frac{3}{5} (e^t + 1) \sqrt[3]{(e^t + 1)^2} + C$$

- 127** Troba la integral de les funcions següents.

$$f(x) = b^x \ln b \cos x - b^x \sin x$$

$$g(x) = b^x \ln b \sin x + b^x \cos x$$

$$\int f(x) dx = b^x \cos x + C$$

$$\int g(x) dx = b^x \sin x + C$$

- 128 Una partícula es desplaça descrivint una espiral logarítmica. Les coordenades de la velocitat són $(b^t \ln b \cos t - b^t \sin t, b^t \ln b \sin t + b^t \cos t)$. Troba les coordenades del vector posició.

Calculem la integral de cada una de les coordenades del vector velocitat.

$$\int b^t \ln b \cos t - b^t \sin t dt = b^t \cos t + C$$

$$\int b^t \ln b \sin t + b^t \cos t dt = b^t \sin t + C$$

$$p(t) = (b^t \cos t, b^t \sin t)$$

- 129 Donada la funció:

$$f(x) = -ab \cos(c - bx)$$

Determina la integral respecte de x , considerant que a, b i c són constants i $F(0) = 0$ quan $c = 0$.

$$F(x) = \int f(x) dx = a \cdot \sin(c - bx) + C$$

$$F(0) = a \cdot \sin c + C = 0 \xrightarrow{c=0} C = -a$$

$$F(x) = a \cdot \sin(c - bx) - a$$

- 130 L'equació de la velocitat d'una ona harmònica és $v(x, t) = \omega A \cos(kx - \omega t)$, en què A és l'ampada, k el nombre d'ona, x la distància des del punt estudiat fins a l'origen de l'ona, ω la freqüència angular i t l'instant de temps. Calcula l'elongació, $y(x, t)$, sabent que la velocitat és la derivada d'aquesta elongació respecte del temps i que $y(0, 0) = 0$.

$$y(x, t) = \int v(x, t) dt = -A \cdot \sin(kx - \omega t) + C$$

$$y(0, 0) = -A \cdot \sin 0 + C = 0 \rightarrow$$

$$\rightarrow -A + C = 0 \rightarrow C = A$$

$$y(x, t) = A \cdot \sin(kx - \omega t) + A$$

Si en vengués 30 parells, obtindria 1900 €.

$$R(30) = 2300 - (30 - 50)^2 = 1900$$

Si en vengués 25 parells, obtindria 1675 €.

$$R(25) = 2300 - (25 - 50)^2 = 1675$$

- 2 Què passa si es venen més de 50 productes? I si se'n venen menys?

Amb la venda de 50 parells de sabatilles esportives s'obté el benefici màxim, i per tant, si els preus no varien, els beneficis començarien a disminuir.

Si se'n venen menys de 50 parells, l'empresa obté beneficis, però no arriba al benefici màxim.

- 3 Defineix l'interval de sabatilles amb què l'empresa obté beneficis.

$$R(x) = 0 \rightarrow 2300 - (x - 50)^2 = 0 \rightarrow$$

$$\rightarrow \begin{cases} x_1 = -10(\sqrt{23} - 5) \approx 2,04 \\ x_2 = 10(5 + \sqrt{23}) \approx 97,96 \end{cases}$$

La funció de benefici s'anul·la en $x = 2,04$

i en $x = 97,96$. Comprovem que en valors intermedis la funció és positiva, prenent, per exemple, $x = 10$.

$$R(10) = 700 > 0$$

Tenim, per tant, que la funció de benefici pren valors positius en l'interval $(2,04; 97,96)$, però, com que estem treballant amb parells de sabates, els valors han de ser enters, per això direm que l'empresa obté beneficis en l'interval $[3, 97]$.

- 4 En aquest exemple es pot trobar el mínim benefici? Recorda que un benefici negatiu s'anomena *pèrdua*.

El benefici mínim s'obté en algun dels extrems de l'interval.

$$R(3) = 91 = R(97)$$

En ambdós extrems s'obté el mateix benefici, 91 €.

MATEMÀTIQUES EN EL MÓN REAL

- 1 Quin benefici obtindrà l'empresa si es venen 30 parells de sabatilles? I si se'n venen 25?

El benefici ve donat per aquesta funció:

$$R(x) = 2300 - (x - 50)^2.$$